

User Manual

for S32 ETH Driver

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Chapter 1

Revision History

Revision	Date	Author	Description
1.0	30.06.2023	NXP RTD Team	S32 Real-Time Drivers AUTOSAR 4.4 Version 4.0.2 Release

Chapter 2

Introduction

- [Supported Derivatives](#)
- [Overview](#)
- [About This Manual](#)
- [Acronyms and Definitions](#)
- [Reference List](#)

This User Manual describes NXP Semiconductors' AUTOSAR Ethernet Driver for S32.

AUTOSAR Ethernet Driver configuration parameters description can be found in the Tressos Configuration Plugin section. Deviations from the specification are described in the [Deviations from Requirements](#) section.

AUTOSAR Ethernet driver requirements and APIs are described in the Ethernet Driver Software Specification Document (version 4.4.0).

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32g274a_bga525
- s32g254a_bga525
- s32g233a_bga525
- s32g234m_bga525
- s32g378a_bga525
- s32g379a_bga525
- s32g398a_bga525
- s32g399a_bga525

- s32g338m_bga525
- s32g339m_bga525
- s32g358a_bga525
- s32g359a_bga525
- s32r45_bga780

All of the above microcontroller devices are collectively named as S32.

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

2.4 Acronyms and Definitions

Term	Definition
API	Application Programming Interface
AUTOSAR	Automotive Open System Architecture
DEM	Diagnostic Event Manager
DET	Default Error Tracer
ETH	Ethernet
ETHIF	Ethernet Interface
ETHTRCV	Ethernet Transceiver
ETHSWT	Ethernet Switch
MCU	Micro controller Unit
MII	Media Independent Interface
N/A	Not Available
RMII	Reduced Media Independent Interface
RAM	Random Access Memory

- The term "Ethernet Controller" is related to the hardware module providing the Ethernet functionality.
- The term "Ethernet Driver" is related to the software handling the Ethernet Controller.
- The term "Application" is used for the software utilizing the Ethernet Driver.

2.5 Reference List

#	Title	Version
1	Specification of Ethernet Driver	AUTOSAR Release 4.↵ 4.0
2	S32G2 Reference Manual	Rev. 7, February 2023
3	S32G3 Reference Manual	Rev. 3, Feb 2023
4	S32R45 Reference Manual	Rev. 3, 12/2021
5	S32G2: Mask Set Errata for Mask 0P77B	Rev. 3.0
6	S32G2: Mask Set Errata for Mask 1P77B	Rev. 1.0
7	S32G3: Mask Set Errata for Mask 1P72B	Rev. 2.1
8	S32R45: Mask Set Errata for Mask P57D	Rev. 2.0
9	S32G2 Data Sheet	Rev 6, Dec 2022
10	S32G3 Data Sheet	Rev 2, RC, Feb 2023
11	VR5510 Data Sheet	Rev 5, April 2022
12	S32R45 Data Sheet	Rev. 3, RC — 11/2022

Chapter 3

Driver

- [Requirements](#)
- [Driver Design Summary](#)
- [Hardware Resources](#)
- [Deviations from Requirements](#)
- [Driver Limitations](#)
- [Driver usage and configuration tips](#)
- [Runtime errors](#)
- [Symbolic Names Disclaimer](#)

3.1 Requirements

Requirements for this driver are detailed in the AUTOSAR 4.4.0 Ethernet Driver Software Specification document (See Table [Reference List](#))

3.2 Driver Design Summary

The Ethernet Driver controls the Gigabit Ethernet Media Access Controller (GMAC) module of the S32 device. It provides the following features:

- Configuration and initialization of the Ethernet Controller
- Switching the Ethernet Controller on and off
- Reception and transmission of Ethernet frames
- Access to the Ethernet Controller's RMON counters (RxStats, TxStats and TxErrorCounterValues)
- Access to the Ethernet Transceiver device registers through MDIO
- Handling of the Ethernet Controller's interrupt requests

- Half and full duplex operation support
- Various PHY Interfaces (10/100 Mbps on MII and RMII, 100/1000 Mbps on RGMII, 1000/2500 Mbps on SGMII)
- Time Synchronization over Ethernet (The gPTP stack has to be provided by the upper layers)
- Time Aware shaper, frame preemption for time sensitive networking.
- Checksum hardware offloading for IPv4/IPv6 with TCP, UDP or ICMP

3.3 Hardware Resources

For the Ethernet driver, there are no differences in hardware resources between derivatives.

The hardware module configured by the Ethernet driver is GMAC (Gigabit Ethernet Media Access Controller). It has a single hardware instance (GMAC_0) for S32G2xx, S32G3xx and 2 instances (GMAC_0 and GMAC_1) for S32R45.

The Ethernet controller to microcontroller pin mapping can be done using the file **S32G2_IOMUX.xlsx** and **S32R45_IOMUX.xlsx** **S32G3_IOMUX.xlsx** attached to the Reference Manual.

3.4 Deviations from Requirements

The driver deviates from the AUTOSAR ETH Driver software specification in some places. The table below identifies the AUTOSAR requirements that are not implemented or out of scope for the ETH Driver.

Term	Definition
N/S	Out of scope
N/I	Not implemented
N/F	Not fully implemented

Below table identifies the AUTOSAR requirements that are not fully implemented, implemented differently or out of scope for the ETH driver.

Requirement	Status	Description	Notes
ECUC_Eth_00035	N/S	Name - EthGetDropCountApi - Parent Container - EthGeneral - Description - Enables / Disables Eth↔_GetCounterValues API. - Multiplicity - 1 - Type - EcucBooleanParamDef - Default value - - - Post-↔Build Variant Value - false - Value Configuration Class - Pre-compile time - X - All Variants - Link time - - - - Post-build time - - - - Scope / Dependency - scope: local -	Just exist in ASR4.2.X

Requirement	Status	Description	Notes
ECUC_Eth_00036	N/S	Name - EthGetEtherStatsApi - Parent Container - EthGeneral - Description - Enables / Disables Eth_↔GetEtherStats API. - Multiplicity - 1 - Type - Ecuc↔BooleanParamDef - Default value - - - Post-Build Variant Value - false - Value Configuration Class - Pre-compile time - X - All Variants - Link time - - - Post-build time - - - Scope / Dependency - scope: local -	Just exist in ASR4.2.X

3.5 Driver Limitations

The Ethernet Driver has the following limitations:

- The API functions *Eth_ReadMii* and *Eth_WriteMii* (and their counterparts *Eth_ReadMmd* and *Eth_Write↔Mmd*) are synchronous instead of asynchronous due to a variety of reasons further detailed in [this](#) AUTOSAR ticket.
- The length of a single received frame (including the 14-bytes Ethernet frame header and the 4-bytes FCS) must be less than or equal to `EthCtrlConfigIngressFifoBufLenByte`.
- When preemption enabled, MMC can incorrectly operate for Tx when there are so many intermediate fragment frames that `SMD-Sx + MCRC_0 + SMD_Cx_1 + FragCount_1 + MCRC_1 + ... + SMD_Cx_n + Frag↔Count_n + FCS` would make the frame transmission time to exceed the jabber limit (i.e. the time limit to transmit an uninterrupted/unfragmented frame with the maximum configured MTU). This is due to Errata ERR050597 can be find in Errata documentation in [<Reference List section>](#).
- VLAN Rx Hash filtering modes weren't supported. Setting bits ETV (Enable VLAN Tag comparison), D↔OVLTC (Disable VLAN Type Check), or ERSVLM (Enable Receive S-VLAN Match) are not available. Use the alternative filtering methods available in the hardware (perfect or inverse VLAN filtering). This is due to Errata ERR050109 can be find in Errata documentation in [<Reference List section>](#).
- Never call the function `Gmac_Ip_SetTxThreshold` when using Time Aware Shaper. This is due to Errata ERR050111 can be find in Errata documentation in [<Reference List section>](#)
- Errata ERR050706 - Received packet would be discarded if preamble byte doesn't precede the SFD,SMD-S or SMD-C byte when frame preemption is enabled. This happens because start of packet detection logic of MAC receiver incorrectly checks the preamble byte. The packets without Preamble or with corrupted Preamble byte before SFD is mostly an error case which has very low probability of occurrence. For the ERR050706 there is no software workaround. This means the applications that support packet re-transmission can re-transmit the packet. This can be find in Errata documentation in [<Reference List section>](#)

3.6 Driver usage and configuration tips

- In case of using only `Gmac_Ip` layer, after invoking [Gmac_Ip_InitSysTime\(\)](#) API to enable ingress/egress timestamping (`MAC_TIMESTAMP_CONTROL[TSENA] = 1`), user must ensure that [Gmac_Ip_ReadFrame\(\)](#) API is always provided a valid `*Info` pointer, as all Rx frames are automatically timestamped.
- For ETH driver, the number of Tx and Rx ring descriptors set by the configuration nodes `EthCtrlConfig↔EgressFifoBufTotal` and `EthCtrlConfigIngressFifoBufTotal` is only limited to the 1024 since this is the range of the Descriptor Ring Length Register. However, by default, the number of descriptors configured on each ring for Tx and Rx is limited to the system memory space. ETH driver allows any number of descriptors to be configured (up to the hardware limitation), but the data buffers allocation must fit into the system memory, otherwise a linker error will occur in the build phase.

3.7 Runtime errors

The Ethernet driver generates the following DEM extended production errors at runtime:

3.7.1 Standardized Extended Production Errors

Function	Error code	Condition triggering the error
Eth_MainFunction	ETH_E_ACCESS	Controller access failed
Eth_MainFunction	ETH_E_RX_FRAMES_LOST	Rx frame lost detected
Eth_MainFunction	ETH_E_CRC	CRC failure detected
Eth_MainFunction	ETH_E_UNDERSIZEFRAME	Undersized frame detected
Eth_MainFunction	ETH_E_OVERSIZEFRAME	Oversized frame detected
Eth_MainFunction	ETH_E_ALIGNMENT	Alignment error detected
Eth_MainFunction	ETH_E_SINGLECOLLISION	Single collision detected
Eth_MainFunction	ETH_E_MULTIPLECOLLISION	Multiple collision detected
Eth_MainFunction	ETH_E_LATECOLLISION	Late collision detected

3.7.2 Vendor-Specific Extended Production Errors

None

3.8 Symbolic Names Disclaimer

All containers having symbolicNameValue set to TRUE in the AUTOSAR schema will generate defines like:

```
#define <Mip>Conf_<Container_ShortName>_<Container_ID>
```

For this reason it is forbidden to duplicate the names of such containers across the RTD configurations or to use names that may trigger other compile issues (e.g. match existing `#ifdefs` arguments).

Chapter 4

Tresos Configuration Plug-in

This chapter describes the Tresos configuration plug-in for the driver. All the parameters are described below.

- Module [Eth](#)
 - Container [EthGeneral](#)
 - * Parameter [EthDevErrorDetect](#)
 - * Parameter [EthGetDropCountApi](#)
 - * Parameter [EthGetEtherStatsApi](#)
 - * Parameter [EthGetCounterValuesApi](#)
 - * Parameter [EthGetRxStatsApi](#)
 - * Parameter [EthGetTxStatsApi](#)
 - * Parameter [EthGetTxErrorCounterValuesApi](#)
 - * Parameter [EthGlobalTimeSupport](#)
 - * Parameter [EthVersionInfoApi](#)
 - * Parameter [EthIndex](#)
 - * Parameter [EthMainFunctionPeriod](#)
 - * Parameter [EthMaxCtrlsSupported](#)
 - * Reference [EthEcucPartitionRef](#)
 - * Container [EthCtrlOffloading](#)
 - Parameter [EthCtrlEnableOffloadChecksumICMP](#)
 - Parameter [EthCtrlEnableOffloadChecksumIPv4](#)
 - Parameter [EthCtrlEnableOffloadChecksumTCP](#)
 - Parameter [EthCtrlEnableOffloadChecksumUDP](#)
 - * Container [EthGeneralVendorSpecific](#)
 - Parameter [EthDisableDemEventDetect](#)
 - Parameter [EthEnableUserModeSupport](#)
 - Parameter [EthMulticoreSupport](#)
 - Parameter [EthUpdatePhysAddrFilterApi](#)
 - Parameter [EthSwtManagementSupportApi](#)
 - Parameter [EthTimeoutMethod](#)
 - Parameter [EthTimeoutDuration](#)
 - Container [EthConfigSet](#)
 - * Container [EthCtrlConfig](#)

- Parameter [EthCtrlEnableMii](#)
- Parameter [EthCtrlEnableMmd](#)
- Parameter [EthCtrlEnableRxInterrupt](#)
- Parameter [EthCtrlEnableTxInterrupt](#)
- Parameter [EthCtrlIdx](#)
- Parameter [EthCtrlMacLayerType](#)
- Parameter [EthCtrlMacLayerSubType](#)
- Parameter [EthCtrlMacLayerSpeed](#)
- Parameter [EthCtrlPhyAddress](#)
- Reference [EthCtrlEcucPartitionRef](#)
- Container [EthCtrlVendorSpecific](#)
- Container [EthCtrlConfigGeneral](#)
- Parameter [EthCtrlEnableAtInit](#)
- Parameter [EthCtrlAllocateTxDataBuffers](#)
- Parameter [EthCtrlAllocateRxDataBuffers](#)
- Parameter [EthEthTrcvDriverVendorIdAndApiInfix](#)
- Parameter [EthEthSwtDriverVendorIdAndApiInfix](#)
- Parameter [EthDuplexMode](#)
- Parameter [EthTimeStampRequiredAccuracy](#)
- Reference [EthModuleReferenceClock](#)
- Reference [EthTimeStampReferenceClock](#)
- Container [EthCtrlConfigMac](#)
- Parameter [MAC_CONFIG_CRC_STRIPPING](#)
- Parameter [MAC_CONFIG_AUTO_PAD](#)
- Parameter [MAC_CONFIG_JUMBO_PKT_EN](#)
- Parameter [MAC_CONFIG_LOOPBACK](#)
- Parameter [MAC_CONFIG_ENABLE_CRS_FD](#)
- Parameter [MAC_CONFIG_DISABLE_RECEIVE_OWN_HD](#)
- Parameter [MAC_CONFIG_DISABLE_CRS_HD](#)
- Parameter [MAC_CONFIG_DISABLE_RETRY](#)
- Parameter [MAC_CONFIG_DEFERRAL_CHECK_HD](#)
- Container [EthCtrlConfigPacketFilter](#)
- Parameter [PKT_FILTER_RECV_ALL](#)
- Parameter [PKT_FILTER_HASH_OR_PERFECT_FILTER](#)
- Parameter [PKT_FILTER_BLOCK_CONTROL_PKTS](#)
- Parameter [PKT_FILTER_BLOCK_PAUSE_PKTS](#)
- Parameter [PKT_FILTER_PASS_ALL_CONTROL_PKTS](#)
- Parameter [PKT_FILTER_PASS_CONTROL_PKTS_ADDR_MATCH](#)
- Parameter [PKT_FILTER_DISABLE_BROADCAST](#)
- Parameter [PKT_FILTER_PASS_ALL_MULTICAST](#)
- Parameter [PKT_FILTER_DST_ADDR_INV_FILTER_EN](#)
- Parameter [PKT_FILTER_HASH_MULTICAST](#)
- Parameter [PKT_FILTER_HASH_UNICAST](#)
- Parameter [PKT_FILTER_PROMISCUOUS_MODE](#)
- Container [EthCtrlConfigEgress](#)
- Reference [EthCtrlConfigEgressLastSchedulerRef](#)
- Container [EthCtrlConfigEgressFifo](#)

- Parameter [EthCtrlConfigEgressFifoBufLenByte](#)
- Parameter [EthCtrlConfigEgressFifoBufTotal](#)
- Parameter [EthCtrlConfigEgressFifoIdx](#)
- Parameter [EthCtrlConfigEgressFifoCallback](#)
- Parameter [EthCtrlConfigEgressFifoPriorityAssignment](#)
- Container [EthCtrlConfigScheduler](#)
- Container [EthCtrlConfigSchedulerPredecessor](#)
- Parameter [EthCtrlConfigSchedulerPredecessorOrder](#)
- Reference [EthCtrlConfigSchedulerPredecessorRef](#)
- Container [EthCtrlConfigSchedulerPredecessorVendorSpecific](#)
- Parameter [EthQueueBandwidth](#)
- Container [EthCtrlConfigSchedulerVendorSpecific](#)
- Parameter [EthTxSchedulerAlgorithm](#)
- Parameter [EthTotalPacketsPerCycle](#)
- Parameter [EthTotalQuantumPerCycle](#)
- Container [EthCtrlConfigShaper](#)
- Parameter [EthCtrlConfigShaperIdleSlope](#)
- Reference [EthCtrlConfigShaperPredecessorFifoRef](#)
- Container [EthCtrlConfigShaperVendorSpecific](#)
- Parameter [EthCtrlConfigHiCredit](#)
- Parameter [EthCtrlConfigLoCredit](#)
- Container [EthCtrlConfigTimeAwareShaper](#)
- Container [EthCtrlConfigTimeAwareShaper](#)
- Parameter [EthGateConfigBaseTimeInSecond](#)
- Parameter [EthGateConfigBaseTimeInNanoSecond](#)
- Parameter [EthGateConfigCycleTime](#)
- Parameter [EthGateConfigExtendTime](#)
- Parameter [EthCtrlEnablePreemption](#)
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- Parameter [EthCtrlConfigInterval](#)
- Container [EthGateControlFifo](#)
- Parameter [EthCtrlConfigGateStatus](#)
- Reference [EthCtrlConfigFifoRef](#)
- Container [EthPreemptionClassification](#)
- Parameter [EthCtrlConfigPreemptionClassification](#)
- Reference [EthPreemptionFifoRef](#)
- Container [EthCtrlConfigIngress](#)
- Container [EthCtrlConfigIngressFifo](#)
- Parameter [EthCtrlConfigIngressFifoBufLenByte](#)
- Parameter [EthCtrlConfigIngressFifoBufTotal](#)
- Parameter [EthCtrlConfigIngressFifoIdx](#)
- Parameter [EthCtrlConfigIngressFifoCallback](#)
- Parameter [EthCtrlConfigIngressFifoPriorityAssignment](#)
- Container [EthDemEventParameterRefs](#)
- Reference [ETH_E_ACCESS](#)

- Reference [ETH_E_RX_FRAMES_LOST](#)
- Reference [ETH_E_CRC](#)
- Reference [ETH_E_UNDERSIZEFRAME](#)
- Reference [ETH_E_OVERSIZEFRAME](#)
- Reference [ETH_E_ALIGNMENT](#)
- Reference [ETH_E_SINGLECOLLISION](#)
- Reference [ETH_E_MULTIPLECOLLISION](#)
- Reference [ETH_E_LATECOLLISION](#)
- Container [CommonPublishedInformation](#)
 - * Parameter [ModuleId](#)
 - * Parameter [VendorId](#)
 - * Parameter [VendorApiInfix](#)
 - * Parameter [ArReleaseMajorVersion](#)
 - * Parameter [ArReleaseMinorVersion](#)
 - * Parameter [ArReleaseRevisionVersion](#)
 - * Parameter [SwMajorVersion](#)
 - * Parameter [SwMinorVersion](#)
 - * Parameter [SwPatchVersion](#)

4.1 Module Eth

Configuration of the Eth (Ethernet Driver) module.

Included containers:

- [EthGeneral](#)
- [EthConfigSet](#)
- [CommonPublishedInformation](#)

Property	Value
type	ECUC-MODULE-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantSupport	true
supportedConfigVariants	VARIANT-PRE-COMPILE, VARIANT-LINK-TIME, VARIANT-POST-BUILD

4.2 Container EthGeneral

General configuration of Ethernet Driver module.

Included subcontainers:

- [EthCtrlOffloading](#)
- [EthGeneralVendorSpecific](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.3 Parameter EthDevErrorDetect

Enables / Disables development error detection.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.4 Parameter EthGetDropCountApi

Obsolete and kept only for backwards compatibility. The Eth_GetDropCount API has been replaced with Eth_GetCounterValues since R4.3.1. Use parameter EthGetCounterValuesApi instead.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.5 Parameter EthGetEtherStatsApi

Obsolete and kept only for backwards compatibility. The Eth_GetEtherStats API has been replaced with the Eth_GetTxStats and Eth_GetRxStats APIs since R4.3.1.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.6 Parameter EthGetCounterValuesApi

Enables / Disables Eth_GetCounterValues API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.7 Parameter EthGetRxStatsApi

Enables / Disables Eth_GetRxStats API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.8 Parameter EthGetTxStatsApi

Enables / Disables Eth_GetTxStats API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.9 Parameter EthGetTxErrorCounterValuesApi

Enables / Disables Eth_GetTxErrorCounterValues API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.10 Parameter EthGlobalTimeSupport

Enables/Disables the GlobalTime APIs used amongst others by Global Time Synchronization over Ethernet.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.11 Parameter EthVersionInfoApi

Enables / Disables Eth_GetVersionInfo API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.12 Parameter EthIndex

Specifies the InstanceId of this module instance. If only one instance is present it shall have the Id 0.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.13 Parameter EthMainFunctionPeriod

The period between successive calls to the main function in seconds.

Ethernet driver does not require this information but the BSW scheduler.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0.001
max	65.535
min	0.001

4.14 Parameter EthMaxCtrlsSupported

Limits the total number of supported controllers.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1
max	1
min	1

4.15 Reference EthEcucPartitionRef

Maps the Ethernet driver to zero or multiple ECUC partitions to make the module's API available in this partition.



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The Ethernet driver will operate as an independent instance in each of the partitions.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

4.16 Container EthCtrlOffloading

Configuration of hardware offloading features.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.17 Parameter EthCtrlEnableOffloadChecksumICMP

Enables / Disables hardware offloading for ICMP checksums.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.18 Parameter EthCtrlEnableOffloadChecksumIPv4

Enables / Disables hardware offloading for IPv4 checksums.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.19 Parameter EthCtrlEnableOffloadChecksumTCP

Enables / Disables hardware offloading for TCP checksums.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.20 Parameter EthCtrlEnableOffloadChecksumUDP

Enables / Disables hardware offloading for UDP checksums.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.21 Container EthGeneralVendorSpecific

General vendor-specific configuration of Ethernet Driver.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.22 Parameter EthDisableDemEventDetect

When this option is enabled, no DEM events are reported.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.23 Parameter EthEnableUserModeSupport

When this parameter is enabled, the Eth module will adapt to run from User Mode, with the following measures:

(if applicable) a) configuring REG_PROT for the Eth Controllers so that the registers under protection eth be accessed from user mode by setting UAA bit in REG_PROT_GCR to 1.

(if applicable) b) using 'call trusted function' stubs for all internal function calls that access registers requiring supervisor mode.

(if applicable) c) other module specific measures for more information, please see chapter 5.7 User Mode Support in IM

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.24 Parameter EthMulticoreSupport

Switches multicore support on or off:

False: For all variants, no EcucPartition shall be referenced in EthEcucPartitionRef.

True: For all variants, at least one EcucPartition needs to be referenced in EthEcucPartitionRef.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.25 Parameter EthUpdatePhysAddrFilterApi

Enables/Disables Eth_UpdatePhysAddrFilter API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.26 Parameter EthSwtManagementSupportApi

Enables / Disables the Switch Management APIs.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.27 Parameter EthTimeoutMethod

Configures the timeout method.

Based on this selection a certain timeout method from OsIf will be used in the driver.

Note: If SystemTimer or CustomTimer are selected make sure the corresponding timer is enabled in OsIf General configuration.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	OSIF_COUNTER_DUMMY
literals	['OSIF_COUNTER_SYSTEM', 'OSIF_COUNTER_CUSTOM', 'OSIF_COUNTER_DUMMY']

4.28 Parameter EthTimeoutDuration

The unit of measurement is given in number of microseconds.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1000
max	65535
min	1

4.29 Container EthConfigSet

This container contains the configuration parameters and sub containers of the AUTOSAR Eth module.

Included subcontainers:

- [EthCtrlConfig](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.30 Container EthCtrlConfig

Configuration of the individual controller

Included subcontainers:

- [EthCtrlVendorSpecific](#)
- [EthCtrlConfigEgress](#)
- [EthCtrlConfigTimeAwareShaper](#)
- [EthCtrlConfigIngress](#)
- [EthDemEventParameterRefs](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	255
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.31 Parameter EthCtrlEnableMii

Enables / Disables Media Independent Interface (MII) for transceiver access.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.32 Parameter EthCtrlEnableMmd

Enables / Disables Clause 45 Media Independent Interface (MII) for transceiver access.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.33 Parameter EthCtrlEnableRxInterrupt

Enables / Disables receive interrupt.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.34 Parameter EthCtrlEnableTxInterrupt

Enables / Disables transmit interrupt.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.35 Parameter EthCtrlIdx

Specifies the instance ID of the configured controller.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	0
min	0

4.36 Parameter EthCtrlMacLayerType

Defines the MAC layer type of the ethernet controller.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	ETH_MAC_LAYER_TYPE_XMII
literals	['ETH_MAC_LAYER_TYPE_XMII', 'ETH_MAC_LAYER_TYPE_XGMII', 'ETH_MAC_LAYER_TYPE_XXGMII']

4.37 Parameter EthCtrlMacLayerSubType

Defines the MAC layer subtype of the ethernet controller.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	STANDARD
literals	['REDUCED', 'SERIAL', 'STANDARD']

4.38 Parameter EthCtrlMacLayerSpeed

Defines the baud rate of the MAC layer.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	ETH_MAC_LAYER_SPEED_10M
literals	['ETH_MAC_LAYER_SPEED_10M', 'ETH_MAC_LAYER_SPEED_100M', 'ETH_MAC_LAYER_SPEED_1G', 'ETH_MAC_LAYER_SPEED_10G', 'ETH_MAC_LAYER_SPEED_2500M']

4.39 Parameter EthCtrlPhyAddress

Specifies the unique 48-bit physical address (MAC address) of the controller in network byte order.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	66:55:44:33:22:11

4.40 Reference EthCtrlEcucPartitionRef

Maps the Ethernet controller to zero or one ECUC partition.

The ECUC partition referenced is a subset of the ECUC partitions where the Ethernet driver is mapped to.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

4.41 Container EthCtrlVendorSpecific

Vendor specific configuration of the individual controller

Included subcontainers:

- [EthCtrlConfigGeneral](#)
- [EthCtrlConfigMac](#)
- [EthCtrlConfigPacketFilter](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.42 Container EthCtrlConfigGeneral

General Controller Configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.43 Parameter EthCtrlEnableAtInit

Enabled - The controller is enabled at initialization.

Disabled - The controller is left disabled at initialization and must be later enabled by calling Gmac_Ip_EnableController.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.44 Parameter EthCtrlAllocateTxDataBuffers

Enabled - The driver will allocate both Tx Buffer Descriptors and Tx Data Buffers.

Disabled - The driver will only allocate Tx Buffer Descriptors. The Tx Data Buffers will be allocated by the application.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.45 Parameter EthCtrlAllocateRxDataBuffers

Enabled - The driver will allocate both Rx Buffer Descriptors and Rx Data Buffers.

Disabled - The driver will only allocate Rx Buffer Descriptors. The Rx Data Buffers will be allocated by the application.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.46 Parameter EthEthTrcvDriverVendorIdAndApiInfix

The vendor id and vendor api infix for the EthTrcv driver, if any.

This is used to include the correct header file and to point to the EthTrcv callbacks.

Example: 43_PHYGROUPB for a 43_PHYGROUPB driver or leave it blank to include EthTrcv.h

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	

4.47 Parameter EthEthSwtDriverVendorIdAndApiInfix

The vendor id and vendor api infix for the EthSwt driver, if any.

This is used to include the correct header file and to point to the EthSwt APIs.

Example: 43_SJA1110 for a 43_SJA1110 driver or leave it blank to include EthSwt.h

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	

4.48 Parameter EthDuplexMode

Defines the controller's duplex mode.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	ETH_FULL_DUPLEX
literals	['ETH_FULL_DUPLEX', 'ETH_HALF_DUPLEX']

4.49 Parameter EthTimeStampRequiredAccuracy

Value given in nanoseconds.

Represents the minimum required accuracy for this gPTP device.

To ensure there's enough headroom for clock drifts from both the GM and Slave, a maximum drift percentage per host must be assumed.

That is, $\text{TotalDriftPercentage} = \text{MaximumGmDriftPercentage} + \text{MaximumSlaveDriftPercentage}$. Then the following inequality must be true:

$$\text{EthTimeStampReferenceClock} * (1 + \text{TotalDriftPercentage} / 100) > (1 / \text{EthTimeStampRequiredAccuracy}) * 10^9.$$

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	40
max	4294967295
min	1

4.50 Reference EthModuleReferenceClock

Reference to the module clock for the GMAC.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/Mcu/McuModuleConfiguration/McuClockSetting↔ Config/McuClockReferencePoint

4.51 Reference EthTimeStampReferenceClock

Reference to the Timestamp source clock for the GMAC.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/Mcu/McuModuleConfiguration/McuClockSetting↔ Config/McuClockReferencePoint

4.52 Container EthCtrlConfigMac

MAC Controller Configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.53 Parameter MAC_CONFIG_CRC_STRIPPING

The last four bytes (FCS) of all packets of Ether type (type field greater than 1,536) are stripped and dropped before forwarding the packet to the application.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.54 Parameter MAC_CONFIG_AUTO_PAD

The MAC strips the Pad or FCS field on the incoming packets only if the value of the length field is less than 1,536 bytes.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.55 Parameter MAC_CONFIG_JUMBO_PKT_EN

The MAC allows jumbo packets of 9,018 bytes (9,022 bytes for VLAN tagged packets).

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.56 Parameter MAC_CONFIG_LOOPBACK

MAC operates in the loopback mode at GMII or MII. The Rx clock input (clk_rx_i) is required for the loopback to work properly.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.57 Parameter MAC_CONFIG_ENABLE_CRS_FD

The MAC transmitter checks the CRS signal before packet transmission in the full-duplex mode. The MAC starts the transmission only when the CRS signal is low.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.58 Parameter MAC_CONFIG_DISABLE_RECEIVE_OWN_HD

The MAC disables the reception of packets when the gmii_txen_o is asserted in the half-duplex mode.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.59 Parameter MAC_CONFIG_DISABLE_CRS_HD

The MAC transmitter ignores the (G)MII CRS signal during packet transmission in the half-duplex mode. As a result, no errors are generated because of Loss of Carrier or No Carrier during transmission.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.60 Parameter MAC_CONFIG_DISABLE_RETRY

The MAC attempts only one transmission. When a collision occurs on the (G)MII interface, the MAC ignores the current packet transmission and reports a Packet Abort with excessive collision error in the Tx packet status.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.61 Parameter MAC_CONFIG_DEFERRAL_CHECK_HD

The deferral check function is enabled in the MAC. The MAC issues a Packet Abort status, along with the excessive deferral error bit set in the Tx packet status, when the Tx state machine is deferred for more than 24,288 bit times in 10/100 Mbps mode (155,680 bits times in 1000/2500 Mbps operation). Deferral begins when the transmitter is ready to transmit, but it is prevented because of an active carrier sense signal (CRS) on (G)MIL.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.62 Container EthCtrlConfigPacketFilter

Packet Filtering Configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.63 Parameter PKT_FILTER_RECV_ALL

Passes all received packets to the application, irrespective of whether they pass the address filter or not.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.64 Parameter PKT_FILTER_HASH_OR_PERFECT_FILTER

The address filter passes a packet if it matches either the perfect filtering or hash filtering.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK

Property	Value
	VARIANT-POST-BUILD: POST-BUILD
default Value	true

4.65 Parameter PKT_FILTER_BLOCK_CONTROL_PKTS

Blocks all control packets from reaching the application.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.66 Parameter PKT_FILTER_BLOCK_PAUSE_PKTS

Forwards all control packets except Pause packets to the application even if they fail the Address filter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.67 Parameter PKT_FILTER_PASS_ALL_CONTROL_PKTS

Forwards all control packets to the application even if they fail the Address filter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.68 Parameter PKT_FILTER_PASS_CONTROL_PKTS_ADDR_MATCH

Forwards the control packets that pass the Address filter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.69 Parameter PKT_FILTER_DISABLE_BROADCAST

Blocks all incoming broadcast packets.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.70 Parameter PKT_FILTER_PASS_ALL_MULTICAST

All received packets with a multicast destination address are passed.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.71 Parameter PKT_FILTER_DST_ADDR_INV_FILTER_EN

The Address checking operates in inverse filtering mode for the destination address comparison for both unicast and multicast packets.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.72 Parameter PKT_FILTER_HASH_MULTICAST

The destination address filtering of received multicast packets is done according to the hash table.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.73 Parameter PKT_FILTER_HASH_UNICAST

The destination address filtering of received unicast packets is done according to the hash table.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.74 Parameter PKT_FILTER_PROMISCUOUS_MODE

Passes all incoming packets irrespective of the destination or source address.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.75 Container EthCtrlConfigEgress

Controller Configuration Egress: Configuration of one Ethernet controller egress behavior.

Included subcontainers:

- [EthCtrlConfigEgressFifo](#)
- [EthCtrlConfigScheduler](#)
- [EthCtrlConfigShaper](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.76 Reference EthCtrlConfigEgressLastSchedulerRef

Reference to the scheduler which is the last in the Egress structure.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfig↵ Egress/EthCtrlConfigScheduler

4.77 Container EthCtrlConfigEgressFifo

Controller Configuration Egress: Represents a FIFO at the egress side.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	5
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.78 Parameter EthCtrlConfigEgressFifoBufLenByte

Length of FIFO elements in bytes.

Note that this represents the length of a single packet buffer (including the 14-bytes Ethernet frame header and the 4-bytes FCS).

Therefore, the payload length will be equal to EthCtrlConfigEgressFifoBufLenByte - 18.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	64
max	1536
min	64

4.79 Parameter EthCtrlConfigEgressFifoBufTotal

FIFO Buffer Count (defines the number of packet buffers of length EthCtrlConfigEgressFifoBufLenByte).

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	16
max	1024
min	4

4.80 Parameter EthCtrlConfigEgressFifoIdx

Egress FIFO index.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	0
max	4
min	0

4.81 Parameter EthCtrlConfigEgressFifoCallback

Callback function invoked when an Egress (Tx) FIFO event is encountered.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	Eth_TxIrqCallback

4.82 Parameter EthCtrlConfigEgressFifoPriorityAssignment

Message egress priority assignment.

Property	Value
type	ECUC-INTEGGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	7
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	0
max	7
min	0

4.83 Container EthCtrlConfigScheduler

Controller Configuration Scheduler: Represents a Scheduler on the Egress side.

Included subcontainers:

- [EthCtrlConfigSchedulerPredecessor](#)
- [EthCtrlConfigSchedulerVendorSpecific](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	255
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.84 Container EthCtrlConfigSchedulerPredecessor

Controller Configuration Scheduler Predecessors: Defines an ordered list of predecessors for this scheduler.

Included subcontainers:

- [EthCtrlConfigSchedulerPredecessorVendorSpecific](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.85 Parameter EthCtrlConfigSchedulerPredecessorOrder

Defines the order of the scheduler predecessors.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.86 Reference EthCtrlConfigSchedulerPredecessorRef

Choice reference to the scheduler predecessor.

Property	Value
type	ECUC-CHOICE-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destinations	['/AUTOSAR/EcuDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfig↵ Egress/EthCtrlConfigEgressFifo', '/AUTOSAR/EcuDefs/Eth/EthConfigSet/↵ EthCtrlConfig/EthCtrlConfigEgress/EthCtrlConfigScheduler', '/AUTOSA↵ R/EcuDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfigEgress/EthCtrl↵ ConfigShaper']

4.87 Container EthCtrlConfigSchedulerPredecessorVendorSpecific

Scheduler Predecessor vendor specific configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.88 Parameter EthQueueBandwidth

Represents the bandwidth percentage allocated for this queue.

In SP, this parameter is irrelevant.

In WRR, $\text{EthQueueBandwidth} / 100 * \text{EthTotalPacketsPerCycle}$ must not exceed 100 packets.

In DWRR, $\text{EthQueueBandwidth} / 100 * \text{EthTotalQuantumPerCycle}$ must not exceed 1,250,000 bytes.

This field is editable only when the scheduling algorithm is set to anything other than STRICT_PRIORITY.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	100
max	100
min	1

4.89 Container EthCtrlConfigSchedulerVendorSpecific

Scheduler vendor specific configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.90 Parameter EthTxSchedulerAlgorithm

Configures the global Tx Scheduling Algorithm affecting all enabled Tx Queues.

For both round-robin algorithms, the queues are serviced in $Q_n > Q_{n-1} > \dots > Q_0$ order based on their allocated bandwidths.

When Credit-Based Shapers are configured in EthCtrlConfigSchedulerPredecessor, the scheduling algorithm must be set to STRICT_PRIORITY.

In a single Egress FIFO configuration, the scheduling algorithm must be set to STRICT_PRIORITY.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	STRICT_PRIORITY
literals	['STRICT_PRIORITY', 'WEIGHTED_ROUND_ROBIN', 'WEIGHTED_FAIR_QUEUING', 'DEFICIT_WEIGHTED_ROUND_ROBIN']

4.91 Parameter EthTotalPacketsPerCycle

Configures the total weight value (in packets) per arbitration cycle from which individual queue weights are calculated based on their allocated bandwidths.

For example, if queues (Q2, Q1, Q0) are configured with bandwidths (50%, 25%, 25%) and EthTotalPacketsPerCycle = 4 packets,

then the individual queue weights are $(4 * 0.5, 4 * 0.25, 4 * 0.25) = (2, 1, 1)$ packets.

This field is editable only when the scheduling algorithm is set to WEIGHTED_ROUND_ROBIN.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	10
max	50000000
min	1

4.92 Parameter EthTotalQuantumPerCycle

Configures the total quantum value (in bytes) per arbitration cycle from which individual queue quantums are calculated based on their allocated bandwidths.

For example, if queues (Q2, Q1, Q0) are configured with bandwidths (50%, 25%, 25%) and EthTotalQuantumPerCycle = 2000 bytes,

then the individual queue quantums are $(2000 * 0.5, 2000 * 0.25, 2000 * 0.25) = (1000, 500, 500)$ bytes.

This field is editable only when the scheduling algorithm is set to DEFICIT_WEIGHTED_ROUND_ROBIN.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1000
max	4294967295
min	1

4.93 Container EthCtrlConfigShaper

Controller Configuration Shaper : Represents a Shaper at the Egress side.

Included subcontainers:

- [EthCtrlConfigShaperVendorSpecific](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	5
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.94 Parameter EthCtrlConfigShaperIdleSlope

Defines the increase of credit in bits per second for the AVB shaper.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	750000000
min	0

4.95 Reference EthCtrlConfigShaperPredecessorFifoRef

Reference to the FIFO which is the predecessor for this Shaper.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfig↔Egress/EthCtrlConfigEgressFifo

4.96 Container EthCtrlConfigShaperVendorSpecific

Shaper vendor specific configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.97 Parameter EthCtrlConfigHiCredit

Defines the maximum value (in bits) that can be accumulated in the credit parameter.

This value should be equal to the maximum size (in bits) of an interfering (best-effort) frame.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	12000
max	131072
min	0

4.98 Parameter EthCtrlConfigLoCredit

Defines the minimum value (in bits) that can be accumulated in the credit parameter.

This value should be equal to twice the maximum size (in bits) of a frame transmitted from this queue.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	4096
max	131072
min	0

4.99 Container EthCtrlConfigTimeAwareShaper

Controller Time Aware Shaper: Represents Time Aware Shaper configuration on the Egress side.

Included subcontainers:

- [EthCtrlConfigTimeAwareShaper](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.100 Container EthCtrlConfigTimeAwareShaper

Controller Time Aware Shaper: Represents Time Aware Shaper configuration on the Egress side.

Included subcontainers:

- [EthGateControlList](#)
- [EthPreemptionClassification](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.101 Parameter EthGateConfigBaseTimeInSecond

The start time for the Gate control list

This value in second.

Once the execution of a given list begins, the implementation can update this value to indicate the next list execution begin time.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1000
max	4294967295
min	0

4.102 Parameter EthGateConfigBaseTimeInNanoSecond

The start time for the Gate control list

This value in nanosecond.

Once the execution of a given list begins, the implementation can update this value to indicate the next list execution begin time.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1000
max	4294967295
min	0

4.103 Parameter EthGateConfigCycleTime

The time at which the execution of the gate control list should be repeated.

This value in nanosecond.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	259294967295
min	0

4.104 Parameter EthGateConfigExtendTime

The parameter holds the amount of time (in nanoseconds) can be extended before switching to the new Gate Control List.

This will be useful in avoiding small fragments of the current list being executed before switching to a new list.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	2147483647
min	0

4.105 Parameter EthCtrlEnablePreemption

Enabled - Frame preemption is enabled.

Disabled - Frame preemption is disabled.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.106 Parameter EthReleaseAdvanceTime

The maximum time in nanoseconds elapse between issuing a RELEASE to the MAC and the MAC being ready to resume

transmission of preemptable frames.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1000
max	65535
min	1

4.107 Parameter EthHoldAdvanceTime

The maximum time in nanoseconds elapse between issuing a HOLD to the MAC and the MAC ceasing to transmit any preemptable frame.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1000
max	65535
min	1

4.108 Container EthGateControlList

Gate Control List : governs the gate close or open events based on schedule provided.

Included subcontainers:

- [EthGateControlFifo](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	256
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.109 Parameter EthCtrlConfigInterval

The time in nano seconds for gate controls are valid and should be applied before reading the next gate controls from the list.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	16000000
min	0

4.110 Container EthGateControlFifo

Gate control fifo: Configure Fifo is open or close in duration interval.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.111 Parameter EthCtrlConfigGateStatus

This parameters to select gate status is open or close.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	OPEN
literals	['CLOSE', 'OPEN']

4.112 Reference EthCtrlConfigFifoRef

Reference to the FIFO which is the predecessor for this time aware shaper.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfig↔Egress/EthCtrlConfigEgressFifo

4.113 Container EthPreemptionClassification

Preemption classification : select reference fifo is preemptable or express.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	5
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.114 Parameter EthCtrlConfigPreemptionClassification

This parameters to select the queue is preemptable or express

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD

Property	Value
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	EXPRESS
literals	['PREEMPTABLE', 'EXPRESS']

4.115 Reference EthPreemptionFifoRef

Reference to the FIFO which is the predecessor for preemption.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfig↔ Egress/EthCtrlConfigEgressFifo

4.116 Container EthCtrlConfigIngress

Controller Configuration Ingress: Configuration of one Ethernet controller ingress behavior.

Included subcontainers:

- [EthCtrlConfigIngressFifo](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.117 Container EthCtrlConfigIngressFifo

Controller Configuration Ingress : Represents a FIFO at the ingress side.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	5
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.118 Parameter EthCtrlConfigIngressFifoBufLenByte

Length of FIFO elements in bytes.

Note that this represents the length of a single packet buffer (including the 14-bytes Ethernet frame header and the 4-bytes FCS).

Therefore, the payload length will be equal to EthCtrlConfigIngressFifoBufLenByte - 18.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	64
max	1536
min	64

4.119 Parameter EthCtrlConfigIngressFifoBufTotal

FIFO Buffer Count (defines the number of packet buffers of length EthCtrlConfigIngressFifoBufLenByte).

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	16
max	1024
min	4

4.120 Parameter EthCtrlConfigIngressFifoIdx

Ingress FIFO index.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	0
max	4
min	0

4.121 Parameter EthCtrlConfigIngressFifoCallback

Callback function invoked when an Ingress (Rx) FIFO event is encountered.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	Eth_RxIrqCallback

4.122 Parameter EthCtrlConfigIngressFifoPriorityAssignment

Message ingress priority assignment.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	7
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	0
max	7
min	0

4.123 Container EthDemEventParameterRefs

Container for the references to DemEventParameter elements which shall be invoked using the API Dem_ReportErrorStatus in case the corresponding error occurs.

The EventId is taken from the referenced DemEventParameter's DemEventId value.

The standardized errors are provided in the container and can be extended by vendor specific error references.

Container is editable only when EthGeneral/EthGeneralVendorSpecific/EthDisableDemEventDetect = FALSE.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.124 Reference ETH_E_ACCESS

Reference to the DemEventParameter which shall be issued when the error 'Controller access failed' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.125 Reference ETH_E_RX_FRAMES_LOST

Reference to the DemEventParameter which shall be issued when the error 'Rx Frame Lost' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.126 Reference ETH_E_CRC

Reference to the DemEventParameter which shall be issued when the error 'CRC Failure' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.127 Reference ETH_E_UNDERSIZEFRAME

Reference to the DemEventParameter which shall be issued when the error 'Undersized Frame' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.128 Reference ETH_E_OVERSIZEFRAME

Reference to the DemEventParameter which shall be issued when the error 'Oversized Frame' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.129 Reference ETH_E_ALIGNMENT

Reference to the DemEventParameter which shall be issued when the error 'Alignment Error' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.130 Reference ETH_E_SINGLECOLLISION

Reference to the DemEventParameter which shall be issued when the error 'Single Collision' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.131 Reference ETH_E_MULTIPLECOLLISION

Reference to the DemEventParameter which shall be issued when the error 'Multiple Collisions' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.132 Reference ETH_E_LATECOLLISION

Reference to the DemEventParameter which shall be issued when the error 'Late Collisions' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.133 Container CommonPublishedInformation

Common container, aggregated by all modules. It contains published information about vendor and versions.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.134 Parameter ModuleId

Module ID of this module from the BSW Module List.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	88
max	88
min	88

4.135 Parameter VendorId

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	43
max	43
min	43

4.136 Parameter VendorApiInfix

In driver modules which can be instantiated several times on a single ECU,

BSW00347 requires that the name of APIs is extended by the VendorId and a vendor specific name.

This parameter is used to specify the vendor specific name.

In total, the implementation specific name is generated as follows:

[ModuleName]_[VendorId]_[VendorApiInfix][API name from SWS].

E.g. Assuming that the VendorId of the implementor is 123 and the implementer chose a

VendorApiInfix of 'v11r456' an API named Can_Write defined in the SWS will translate to Can_123_v11r456Write.

This parameter is mandatory for all modules with upper multiplicity > 1.

It shall not be used for modules with upper multiplicity = 1.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION

4.137 Parameter ArReleaseMajorVersion

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.138 Parameter ArReleaseMinorVersion

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.139 Parameter ArReleaseRevisionVersion

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.140 Parameter SwMajorVersion

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.141 Parameter SwMinorVersion

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.142 Parameter SwPatchVersion

Patch version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	2
max	2
min	2



Chapter 5

Module Index

5.1 Software Specification

Here is a list of all modules:

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Chapter 6

Data Structure Index

6.1 Data Structures

Here are the data structures with brief descriptions:

Gmac_Ip_ManagementInfo	
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Chapter 7

Module Documentation

7.1 ETH Driver

7.1.1 Detailed Description

7.1.1.1 Functions Expectation called for transmission in polling mode:

- Eth_ProvideTxBuffer
- Eth_Transmit
- Eth_TxConfirmation

Macros

- #define `ETH_SID_INIT`
Service ID of Eth_Init.
- #define `ETH_SID_SETCONTROLLERMODE`
Service ID of Eth_SetControllerMode.
- #define `ETH_SID_GETCONTROLLERMODE`
Service ID of Eth_GetControllerMode.
- #define `ETH_SID_WRITEMII`
Service ID of Eth_WriteMii.
- #define `ETH_SID_READMII`
Service ID of Eth_ReadMii.
- #define `ETH_SID_GETPHYSADDR`
Service ID of Eth_GetPhysAddr.
- #define `ETH_SID_PROVIDETXBUFFER`
Service ID of Eth_ProvideTxBuffer.
- #define `ETH_SID_TRANSMIT`
Service ID of Eth_Transmit.
- #define `ETH_SID_RECEIVE`
Service ID of Eth_Receive.

- #define [ETH_SID_TXCONFIRMATION](#)
Service ID of Eth_TxConfirmation.
- #define [ETH_SID_GETVERSIONINFO](#)
Service ID of Eth_GetVersionInfo.
- #define [ETH_SID_UPDATEADDRFILTER](#)
Service ID of Eth_UpdatePhysAddrFilter.
- #define [ETH_SID_SETPHYSADDR](#)
Service ID of Eth_SetPhysAddr.
- #define [ETH_SID_GETCOUNTERVALUE](#)
Service ID of Eth_GetCounterValues.
- #define [ETH_SID_GETRXSTATS](#)
Service ID of Eth_GetRxStats.
- #define [ETH_SID_GETTXSTATS](#)
Service ID of Eth_GetTxStats.
- #define [ETH_SID_GETTXERRORCOUNTERVALUE](#)
Service ID of Eth_GetTxErrorCounterValues.
- #define [ETH_SID_MAINFUNCTION](#)
Service ID of Eth_MainFunction.
- #define [ETH_SID_WRITEMMD](#)
Service ID of Eth_WriteMmd.
- #define [ETH_SID_READMMD](#)
Service ID of Eth_ReadMmd.
- #define [ETH_SID_TXTIMEAWARESHAPER](#)
Service ID of Eth_TxTimeAwareShaperInit.
- #define [ETH_E_INV_CTRL_IDX](#)
Development Error ID for "Invalid Controller Index".
- #define [ETH_E_UNINIT](#)
Development Error ID for "Uninitialized Ethernet Controller".
- #define [ETH_E_PARAM_POINTER](#)
Development Error ID for "Invalid (Null) Pointer Parameter".
- #define [ETH_E_INV_PARAM](#)
Development Error ID for "Invalid Parameter".
- #define [ETH_E_INV_MODE](#)
Development Error ID for "Invalid Controller Mode".
- #define [ETH_E_PARAM_CONFIG](#)
Development Error ID for "Invalid Core ID".
- #define [ETH_E_INIT_FAILED](#)
Development Error ID for "Invalid Configuration pointer argument".

Function Reference

- void [Eth_Init](#) (const Eth_ConfigType *CfgPtr)
Initializes the Ethernet Driver.
- Std_ReturnType [Eth_SetControllerMode](#) (uint8 CtrlIdx, Eth_ModeType CtrlMode)
Enables or disables the given controller.
- Std_ReturnType [Eth_GetControllerMode](#) (uint8 CtrlIdx, Eth_ModeType *CtrlModePtr)

- Obtains the mode of the given controller.*

 - Std_ReturnType [Eth_TxTimeAwareShaperInit](#) (uint8 CtrlIdx)

Control transmit frame in duration time.
- void [Eth_GetPhysAddr](#) (uint8 CtrlIdx, uint8 *PhysAddrPtr)

Obtains the physical source address used by the indexed controller (the node MAC address).
- void [Eth_SetPhysAddr](#) (uint8 CtrlIdx, const uint8 *PhysAddrPtr)

Set or change physical address to the defined controller.
- Std_ReturnType [Eth_UpdatePhysAddrFilter](#) (uint8 CtrlIdx, const uint8 *PhysAddrPtr, Eth_FilterActionType Action)

Adds or removes the specified PhysAddrPtr address to or from a multicast address pool in the controller specified by CtrlIdx.
- Std_ReturnType [Eth_GetCounterValues](#) (uint8 CtrlIdx, Eth_CounterType *CounterPtr)

Reads a list with drop counter values of the corresponding controller.
- Std_ReturnType [Eth_ReadMmd](#) (uint8 CtrlIdx, uint8 TrcvIdx, uint8 Mmd, uint16 RegAddress, uint16 *RegValPtr)

Transfers management data over the MII.
- Std_ReturnType [Eth_WriteMmd](#) (uint8 CtrlIdx, uint8 TrcvIdx, uint8 Mmd, uint16 RegAddress, uint16 RegVal)

Transfers management data over the MII.
- Std_ReturnType [Eth_WriteMii](#) (uint8 CtrlIdx, uint8 TrcvIdx, uint8 RegIdx, uint16 RegVal)

Writes to a transceiver (physical layer driver) register.
- Std_ReturnType [Eth_ReadMii](#) (uint8 CtrlIdx, uint8 TrcvIdx, uint8 RegIdx, uint16 *RegValPtr)

Reads a transceiver (physical layer driver) register.
- Std_ReturnType [Eth_GetRxStats](#) (uint8 CtrlIdx, Eth_RxStatsType *RxStats)

Read the status of a controller.
- Std_ReturnType [Eth_GetTxStats](#) (uint8 CtrlIdx, Eth_TxStatsType *TxStats)

Return the list of Transmission Statistics.
- Std_ReturnType [Eth_GetTxErrorCounterValues](#) (uint8 CtrlIdx, Eth_TxErrorCounterValuesType *TxErrorCounterValues)

Return the list of Transmission Statistics.
- BufReq_ReturnType [Eth_ProvideTxBuffer](#) (uint8 CtrlIdx, uint8 Priority, Eth_BufIdxType *BufIdxPtr, uint8 **BufPtr, uint16 *LenBytePtr)

Provides access to a transmit buffer of the specified controller.
- Std_ReturnType [Eth_Transmit](#) (uint8 CtrlIdx, Eth_BufIdxType BufIdx, Eth_FrameType FrameType, boolean TxConfirmation, uint16 LenByte, const uint8 *PhysAddrPtr)

Triggers transmission of a previously granted and then filled transmit buffer.
- void [Eth_Receive](#) (uint8 CtrlIdx, uint8 FifoIdx, Eth_RxStatusType *RxStatusPtr)

Triggers frames reception notifications.
- void [Eth_TxConfirmation](#) (uint8 CtrlIdx)

Triggers frame transmission confirmations.
- void [Eth_GetVersionInfo](#) (Std_VersionInfoType *VersionInfoPtr)

Returns the version information of this module.

7.1.2 Macro Definition Documentation

7.1.2.1 ETH_SID_INIT

```
#define ETH_SID_INIT
```

Service ID of Eth_Init.

Definition at line 92 of file Eth.h.

7.1.2.2 ETH_SID_SETCONTROLLERMODE

```
#define ETH_SID_SETCONTROLLERMODE
```

Service ID of Eth_SetControllerMode.

Definition at line 94 of file Eth.h.

7.1.2.3 ETH_SID_GETCONTROLLERMODE

```
#define ETH_SID_GETCONTROLLERMODE
```

Service ID of Eth_GetControllerMode.

Definition at line 96 of file Eth.h.

7.1.2.4 ETH_SID_WRITEMII

```
#define ETH_SID_WRITEMII
```

Service ID of Eth_WriteMii.

Definition at line 100 of file Eth.h.

7.1.2.5 ETH_SID_READMII

```
#define ETH_SID_READMII
```

Service ID of Eth_ReadMii.

Definition at line 102 of file Eth.h.

7.1.2.6 ETH_SID_GETPHYSADDR

```
#define ETH_SID_GETPHYSADDR
```

Service ID of Eth_GetPhysAddr.

Definition at line 106 of file Eth.h.

7.1.2.7 ETH_SID_PROVIDETXBUFFER

```
#define ETH_SID_PROVIDETXBUFFER
```

Service ID of Eth_ProvideTxBuffer.

Definition at line 108 of file Eth.h.

7.1.2.8 ETH_SID_TRANSMIT

```
#define ETH_SID_TRANSMIT
```

Service ID of Eth_Transmit.

Definition at line 110 of file Eth.h.

7.1.2.9 ETH_SID_RECEIVE

```
#define ETH_SID_RECEIVE
```

Service ID of Eth_Receive.

Definition at line 112 of file Eth.h.

7.1.2.10 ETH_SID_TXCONFIRMATION

```
#define ETH_SID_TXCONFIRMATION
```

Service ID of Eth_TxConfirmation.

Definition at line 114 of file Eth.h.

7.1.2.11 ETH_SID_GETVERSIONINFO

```
#define ETH_SID_GETVERSIONINFO
```

Service ID of Eth_GetVersionInfo.

Definition at line 118 of file Eth.h.

7.1.2.12 ETH_SID_UPDATEADDRFILTER

```
#define ETH_SID_UPDATEADDRFILTER
```

Service ID of Eth_UpdatePhysAddrFilter.

Definition at line 123 of file Eth.h.

7.1.2.13 ETH_SID_SETPHYSADDR

```
#define ETH_SID_SETPHYSADDR
```

Service ID of Eth_SetPhysAddr.

Definition at line 127 of file Eth.h.

7.1.2.14 ETH_SID_GETCOUNTERVALUE

```
#define ETH_SID_GETCOUNTERVALUE
```

Service ID of Eth_GetCounterValues.

Definition at line 131 of file Eth.h.

7.1.2.15 ETH_SID_GETRXSTATS

```
#define ETH_SID_GETRXSTATS
```

Service ID of Eth_GetRxStats.

Definition at line 136 of file Eth.h.

7.1.2.16 ETH_SID_GETTXSTATS

```
#define ETH_SID_GETTXSTATS
```

Service ID of Eth_GetTxStats.

Definition at line 156 of file Eth.h.

7.1.2.17 ETH_SID_GETTXERRORCOUNTERVALUE

```
#define ETH_SID_GETTXERRORCOUNTERVALUE
```

Service ID of Eth_GetTxErrorCounterValues.

Definition at line 161 of file Eth.h.

7.1.2.18 ETH_SID_MAINFUNCTION

```
#define ETH_SID_MAINFUNCTION
```

Service ID of Eth_MainFunction.

Definition at line 165 of file Eth.h.

7.1.2.19 ETH_SID_WRITEMMD

```
#define ETH_SID_WRITEMMD
```

Service ID of Eth_WriteMmd.

Definition at line 169 of file Eth.h.

7.1.2.20 ETH_SID_READMMD

```
#define ETH_SID_READMMD
```

Service ID of Eth_ReadMmd.

Definition at line 171 of file Eth.h.

7.1.2.21 ETH_SID_TXTIMEAWARESHAPER

```
#define ETH_SID_TXTIMEAWARESHAPER
```

Service ID of Eth_TxTimeAwareShaperInit.

Definition at line 175 of file Eth.h.

7.1.2.22 ETH_E_INV_CTRL_IDX

```
#define ETH_E_INV_CTRL_IDX
```

Development Error ID for "Invalid Controller Index".

Definition at line 183 of file Eth.h.

7.1.2.23 ETH_E_UNINIT

```
#define ETH_E_UNINIT
```

Development Error ID for "Uninitialized Ethernet Controller".

Definition at line 185 of file Eth.h.

7.1.2.24 ETH_E_PARAM_POINTER

```
#define ETH_E_PARAM_POINTER
```

Development Error ID for "Invalid (Null) Pointer Parameter".

Definition at line 187 of file Eth.h.

7.1.2.25 ETH_E_INV_PARAM

```
#define ETH_E_INV_PARAM
```

Development Error ID for "Invalid Parameter".

Definition at line 189 of file Eth.h.

7.1.2.26 ETH_E_INV_MODE

```
#define ETH_E_INV_MODE
```

Development Error ID for "Invalid Controller Mode".

Definition at line 191 of file Eth.h.

7.1.2.27 ETH_E_PARAM_CONFIG

```
#define ETH_E_PARAM_CONFIG
```

Development Error ID for "Invalid Core ID".

Definition at line 194 of file Eth.h.

7.1.2.28 ETH_E_INIT_FAILED

```
#define ETH_E_INIT_FAILED
```

Development Error ID for "Invalid Configuration pointer argument".

Definition at line 197 of file Eth.h.

7.1.3 Function Reference

7.1.3.1 Eth_Init()

```
void Eth_Init (
    const Eth_ConfigType * CfgPtr )
```

Initializes the Ethernet Driver.

The configuration pointer is internally stored and the driver is initialized. The Ethernet controller is also reset.

Note

Function should be called only once.

Warning

Second call can cause undefined behavior. Call the [Eth_SetControllerMode\(\)](#) and pass ETH_MODE_DOWN to the CtrlMode argument before the second Eth_Init call to avoid problems.

Parameters

in	<i>CfgPtr</i>	Points to the implementation specific structure containing the Eth driver configuration Compiler_Warning: this warning due to behavior of compiler depend on configs. implements Eth_Init_Activity
----	---------------	---

7.1.3.2 Eth_SetControllerMode()

```
Std_ReturnType Eth_SetControllerMode (
    uint8 CtrlIdx,
    Eth_ModeType CtrlMode )
```

Enables or disables the given controller.

Warning

Disabling the controller clears all receive and transmit buffers. The application should ensure that no data is lost.

Parameters

in	<i>CtrlIdx</i>	Index of the controller to be enabled or disabled. The index is valid within the context of the Ethernet Driver only.
in	<i>CtrlMode</i>	Mode which shall be entered • ETH_MODE_DOWN: disable the controller • ETH_MODE_ACTIVE: enable the controller

Returns

Error status

Return values

<i>E_OK</i>	No error was detected during the function execution.
<i>E_NOT_OK</i>	Development error was detected and the function failed. implements Eth_SetControllerMode_Activity

7.1.3.3 Eth_GetControllerMode()

```
Std_ReturnType Eth_GetControllerMode (
```

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```
uint8 CtrlIdx,  
Eth_ModeType * CtrlModePtr )
```

Obtains the mode of the given controller.

Parameters

in	<i>CtrlIdx</i>	Index of the controller which state shall be read. The index is valid within the context of the Ethernet Driver only.
out	<i>CtrlModePtr</i>	Pointer where to store the current controller mode.

Returns

Error status

Return values

<i>E_OK</i>	No error was detected during the function execution.
<i>E_NOT_OK</i>	Development error was detected and the function failed. implements Eth_GetControllerMode_Activity

7.1.3.4 Eth_TxTimeAwareShaperInit()

```
Std_ReturnType Eth_TxTimeAwareShaperInit (  
    uint8 CtrlIdx )
```

Control transmit frame in duration time.

Parameters

in	<i>CtrlIdx</i>	Index of the controller which state shall be read. The index is valid within the context of the Ethernet Driver only.
----	----------------	---

Returns

Std_ReturnType

7.1.3.5 Eth_GetPhysAddr()

```
void Eth_GetPhysAddr (  
    uint8 CtrlIdx,  
    uint8 * PhysAddrPtr )
```

Obtains the physical source address used by the indexed controller (the node MAC address).

Parameters

in	<i>CtrlIdx</i>	Index of the controller whose MAC address should be read. The index is valid within the context of the Ethernet Driver only.
out	<i>PhysAddrPtr</i>	Pointer where to store the physical source address (MAC address). The address in network byte order is stored into 6 bytes at the given memory address. implements Eth_GetPhysAddr_Activity

7.1.3.6 Eth_SetPhysAddr()

```
void Eth_SetPhysAddr (
    uint8 CtrlIdx,
    const uint8 * PhysAddrPtr )
```

Set or change physical address to the defined controller.

Parameters

in	<i>CtrlIdx</i>	Index of the controller whose MAC address should be changed. The index is valid within the context of the Ethernet Driver only.
in	<i>PhysAddrPtr</i>	Pointer to the physical source address which should be set to the controller. The address is stored in 6 bytes of memory in network byte order.

Warning

This function may be called only when the controller is down. Calling function Eth_Init will change the controller's MAC address to the default value! implements Eth_SetPhysAddr_Activity

7.1.3.7 Eth_UpdatePhysAddrFilter()

```
Std_ReturnType Eth_UpdatePhysAddrFilter (
    uint8 CtrlIdx,
    const uint8 * PhysAddrPtr,
    Eth_FilterActionType Action )
```

Adds or removes the specified PhysAddrPtr address to or from a multicast address pool in the controller specified by CtrlIdx.

Enables or disables reception for the specified multicast physical address. Unicast addresses are ignored. Operations for special Physical addresses follow. If Physical Address ff:ff:ff:ff:ff:ff is added into a filter (Action=ETH_ADD_TO_FILTER) the filter is completely open and any address is accepted at reception. Later on when Physical Address ff:ff:ff:ff:ff:ff is removed from the filter (Action=ETH_REMOVE_FROM_FILTER) the filtering is recovered and the

reception is allowed again only for addresses remaining in the filter. If Physical Address 00:00:00:00:00:00 is added into a filter, (Action=ETH_ADD_TO_FILTER) the filter is completely close have no any address is received. Later on when Physical Address 00:00:00:00:00:00 is removed from the filter (Action=ETH_REMOVE_FROM_FILTER) the filtering is recovered and the reception is allowed again only for addresses remaining in the filter. Note that operations of full open or close are in exclusive disjunction. Operation of full open excludes full close and vice versa.

Parameters

in	<i>CtrlIdx</i>	Index of the controller. The index is valid within the context of the Ethernet Driver only.
in	<i>PhysAddrPtr</i>	Pointer to PHY address which shall be added or removed to or from multicast pool. The address in network byte order stored into 6 bytes of memory.
in	<i>Action</i>	Determine whenever the defined address will be added to the pool ETH_ADD_TO_FILTER or removed from it ETH_REMOVE_FROM_FILTER. implements Eth_UpdatePhysAddrFilter_Activity

7.1.3.8 Eth_GetCounterValues()

```
Std_ReturnType Eth_GetCounterValues (
    uint8 CtrlIdx,
    Eth_CounterType * CounterPtr )
```

Reads a list with drop counter values of the corresponding controller.

Parameters

in	<i>CtrlIdx</i>	Index of the controller
out	<i>CounterPtr</i>	Pointer to Eth_CounterType where to store the read drop counter values

Returns

Error status

Return values

<i>E_OK</i>	No error was detected during the function execution.
<i>E_NOT_OK</i>	Development error was detected or inaccessible to counters register and the function.

Reads a list with drop counter values of the corresponding controller. The meaning of these values is hardware dependent. However, the list DropCount[] shall contain the following values in the given order, where the maximal possible value shall denote an invalid value, e.g. if this counter is not available:

1. Dropped packets due to buffer overrun
2. Dropped packets due to CRC errors
3. Number of undersize packets which were less than 64 octets long (excluding framing bits, but including FCS octets) and were otherwise well formed. (see IETF RFC 1757)
4. Number of oversize packets which are longer than 1518 octets (excluding framing bits, but including FCS octets) and were otherwise well formed. (see IETF RFC 1757)

5. Number of alignment errors, i.e. packets which are received and are not an integral number of octets in length and do not pass the CRC.
6. SQE test error according to IETF RFC1643 dot3StatsSQETestErrors
7. The number of inbound packets which were chosen to be discarded even though no errors had been detected to prevent their being deliverable to a higher layer protocol. One possible reason for discarding such a packet could be to free up buffer space. (see IETF RFC 2233 ifInDiscards)
8. Total number of erroneous inbound packets
9. The number of outbound packets which were chosen to be discarded even though no errors had been detected to prevent their being transmitted. One possible reason for discarding such a packet could be to free up buffer space. (see IETF RFC 2233 ifOutDiscards)
10. Total number of erroneous outbound packets
11. Single collision frames: A count of successfully transmitted frames on a particular interface for which transmission is inhibited by exactly one collision. (see IETF RFC1643 dot3StatsSingleCollisionFrames)
12. Multiple collision frames: A count of successfully transmitted frames on a particular interface for which transmission is inhibited by more than one collision. (see IETF RFC1643 dot3StatsMultipleCollisionFrames)
13. Number of deferred transmission: A count of frames for which the first transmission attempt on a particular interface is delayed because the medium is busy. (see IETF RFC1643 dot3StatsDeferredTransmissions)
14. Number of late collisions: The number of times that a collision is detected on a particular interface later than 512 bit times into the transmission of a packet. (see IETF RFC1643 dot3StatsLateCollisions)
15. The following positions in the list can contain hardware dependent counter values implements Eth_GetCounterValues_Activity

7.1.3.9 Eth_ReadMmd()

```
Std_ReturnType Eth_ReadMmd (
    uint8 CtrlIdx,
    uint8 TrcvIdx,
    uint8 Mmd,
    uint16 RegAddress,
    uint16 * RegValPtr )
```

Transfers management data over the MII.

Parameters

in	<i>CtrlIdx</i>	Index of controller to be transferred
in	<i>TrcvIdx</i>	Transceiver/Port to communicate with
in	<i>Mmd</i>	Device which shall be accessed
in	<i>RegAddress</i>	Register address which shall be accessed
out	<i>RegValPtr</i>	Data to be written to register

Returns

Failure status

Return values

<i>E_OK</i>	Success, no error has occurred.
<i>E_NOT_OK</i>	Failure, no response from the hardware within a timeout.

7.1.3.10 Eth_WriteMmd()

```
Std_ReturnType Eth_WriteMmd (
    uint8 CtrlIdx,
    uint8 TrcvIdx,
    uint8 Mmd,
    uint16 RegAddress,
    uint16 RegVal )
```

Transfers management data over the MII.

Parameters

in	<i>CtrlIdx</i>	Index of controller to be transferred
in	<i>TrcvIdx</i>	Transceiver/Port to communicate with
in	<i>Mmd</i>	Device which shall be accessed
in	<i>RegAddress</i>	Register address which shall be accessed
in	<i>RegVal</i>	Data to be written to register

Returns

Failure status

Return values

<i>E_OK</i>	Success, no error has occurred.
<i>E_NOT_OK</i>	Failure, no response from the hardware within a timeout.

7.1.3.11 Eth_WriteMii()

```
Std_ReturnType Eth_WriteMii (
    uint8 CtrlIdx,
```

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```
uint8 TrcvIdx,  
uint8 RegIdx,  
uint16 RegVal )
```

Writes to a transceiver (physical layer driver) register.

The management frame is assembled and the MII bus transaction is issued in order to transfer the data. Function waits until the bus transaction finishes.

Warning

This function is blocking the execution until the MII bus transaction is finished.

Parameters

in	<i>CtrlIdx</i>	Index of the controller which transceiver register shall be written. The index is valid within the context of the Ethernet Driver only.
in	<i>TrcvIdx</i>	Index of the transceiver connected the MII. The value shall be within the range 0..31.
in	<i>RegIdx</i>	Index of the transceiver register to be written. The value shall be withing the range 0..31.
in	<i>RegVal</i>	Value to be written into the indexed register.

Return values

<i>E_OK</i>	No error was detected during the function execution.
<i>E_NOT_OK</i>	Development error or the function failed. implements Eth_WriteMii_Activity

7.1.3.12 Eth_ReadMii()

```
Std_ReturnType Eth_ReadMii (  
    uint8 CtrlIdx,  
    uint8 TrcvIdx,  
    uint8 RegIdx,  
    uint16 * RegValPtr )
```

Reads a transceiver (physical layer driver) register.

Parameters

in	<i>CtrlIdx</i>	Index of the controller which transceiver register shall be read. The index is valid within the context of the Ethernet Driver only.
in	<i>TrcvIdx</i>	Index of the transceiver connected on the MII. The value shall be within the range 0..31.
in	<i>RegIdx</i>	Index of the transceiver register to be read. The Value shall be within the range 0..31.
out	<i>RegValPtr</i>	Filled with the register content of the indexed register

Return values

<i>E_OK</i>	No error was detected during the function execution.
<i>E_NOT_OK</i>	Development error or the function failed.

The management frame is assembled and the MII bus transaction is issued in order to transfer the data. Function waits until the bus transaction finishes and then returns the read data.

Warning

This function is blocking the execution until the MII bus transaction is finished. implements Eth_ReadMii↵_Activity

7.1.3.13 Eth_GetRxStats()

```
Std_ReturnType Eth_GetRxStats (
    uint8 CtrlIdx,
    Eth_RxStatsType * RxStats )
```

Read the status of a controller.

Returns the following list according to IETF RFC2819, where the maximal possible value shall denote an invalid value, e.g. if this counter is not available:

1. etherStatsDropEvents
2. etherStatsOctets
3. etherStatsPkts
4. etherStatsBroadcastPkts
5. etherStatsMulticastPkts
6. etherStatsCrcAlignErrors
7. etherStatsUndersizePkts
8. etherStatsOversizePkts
9. etherStatsFragments
10. etherStatsJabbers
11. etherStatsCollisions
12. etherStatsPkts64Octets
13. etherStatsPkts65to127Octets
14. etherStatsPkts128to255Octets
15. etherStatsPkts256to511Octets
16. etherStatsPkts512to1023Octets
17. etherStatsPkts1024to1518Octets

Module Documentation

Parameters

in	<i>CtrlIdx</i>	Index of the controller which shall be read the status register.
out	<i>RxStats</i>	Pointer to 32 bit long memory space to be filled with the list values according to IETF RFC 2819 (Remote Network Monitoring Management Information Base).

Returns

Error status

Return values

<i>E_OK</i>	No error was detected during the function execution.
<i>E_NOT_OK</i>	Development error was detected or inaccessible to counters register and the function. implements Eth_GetRxStats_Activity

7.1.3.14 Eth_GetTxStats()

```
Std_ReturnType Eth_GetTxStats (
    uint8 CtrlIdx,
    Eth_TxStatsType * TxStats )
```

Return the list of Transmission Statistics.

Parameters

in	<i>CtrlIdx</i>	Index of controller within the context of the Ethernet Driver.
out	<i>TxStats</i>	Pointer to 32 bit long memory space to be filled with the list values according to IETF RFC 1213

Return the list of Transmission Statistics out of IETF RFC 1213 defined within Eth_TxStatsType, where the maximal possible value shall denote an invalid value. e.g. this counter is not available..

Returns

The search status

Return values

<i>E_OK</i>	Success
<i>E_NOT_OK</i>	Tx-statistics could not be obtained. implements Eth_GetTxStats_Activity

7.1.3.15 Eth_GetTxErrorCounterValues()

```
Std_ReturnType Eth_GetTxErrorCounterValues (
    uint8 CtrlIdx,
    Eth_TxErrorCounterValuesType * TxErrorCounterValues )
```

Return the list of Transmission Statistics.

Parameters

in	<i>CtrlIdx</i>	Index of controller within the context of the Ethernet Driver.
out	<i>TxErrorCounterValues</i>	List of values to read statistic error counter values for transmission.

Return the list of Transmission Error Counters out of IETF RFC1213 and RFC1643 defined within Eth_TxErrorCounterValuesType, where the maximal possible value shall denote an invalid value. e.g. this counter is not available.

Returns

The search status

Return values

<i>E_OK</i>	Success
<i>E_NOT_OK</i>	Tx-statistics could not be obtained. implements Eth_GetTxErrorCounterValues_Activity

7.1.3.16 Eth_ProvideTxBuffer()

```
BufReq_ReturnType Eth_ProvideTxBuffer (
    uint8 CtrlIdx,
    uint8 Priority,
    Eth_BufIdxType * BufIdxPtr,
    uint8 ** BufPtr,
    uint16 * LenBytePtr )
```

Provides access to a transmit buffer of the specified controller.

Warning

The application should handle possible difference between the requested and granted buffer lengths. It is not necessary to use whole granted buffer i.e. some space at the end may not be written.

Module Documentation

Parameters

in	<i>CtrlIdx</i>	Index of the controller which buffer shall be provided. The index is valid within the context of the Ethernet Driver only.
in	<i>Priority</i>	Frame priority for transmit buffer FIFO selection
out	<i>BufIdxPtr</i>	Index to the granted transmit buffer resource. It uniquely identifies the buffer in all subsequent calls of functions Eth_Transmit() and Eth_TxConfirmation() .
out	<i>BufPtr</i>	Pointer to the granted buffer. This is the space where the data to be transmitted shall be stored.
in, out	<i>LenBytePtr</i>	Buffer payload length - In: desired length in bytes - Out: granted length in bytes

Returns

Error and buffer status

Return values

<i>BUFREQ_OK</i>	Buffer was successfully granted and no error has occurred.
<i>BUFREQ_E_NOT_OK</i>	A development error was detected and no buffer was granted.
<i>BUFREQ_E_BUSY</i>	All available buffers in use therefore no buffer was granted. No error has been detected. implements Eth_ProvideTxBuffer_Activity

7.1.3.17 Eth_Transmit()

```
Std_ReturnType Eth_Transmit (  
    uint8 CtrlIdx,  
    Eth_BufIdxType BufIdx,  
    Eth_FrameType FrameType,  
    boolean TxConfirmation,  
    uint16 LenByte,  
    const uint8 * PhysAddrPtr )
```

Triggers transmission of a previously granted and then filled transmit buffer.

Parameters

in	<i>CtrlIdx</i>	Index of the controller which buffer shall be transmitted. The index is valid within the context of the Ethernet Driver only.
in	<i>BufIdx</i>	Index of the buffer resource to be transmitted.
in	<i>FrameType</i>	Desired value of the Ethernet frame type in the frame header.
in	<i>TxConfirmation</i>	Activates transmission confirmation.
in	<i>LenByte</i>	Buffer data length in bytes (payload length).
in	<i>PhysAddrPtr</i>	Physical target address (MAC address) in network byte order.

Returns

Error status

Return values

<i>E_OK</i>	No error was detected during the function execution.
<i>E_NOT_OK</i>	Development error was detected and the function failed. implements Eth_Transmit_Activity

7.1.3.18 Eth_Receive()

```
void Eth_Receive (
    uint8 CtrlIdx,
    uint8 FifoIdx,
    Eth_RxStatusType * RxStatusPtr )
```

Triggers frames reception notifications.

All receive buffers are checked and the first received frame is passed to the EthIf module. The caller is notified whether any frame was received and whether more frames are available in the receive queue.

Parameters

in	<i>CtrlIdx</i>	Index of the controller which shall be checked whether any new frames were received. The index is valid within the context of the Ethernet Driver only.
in	<i>FifoIdx</i>	Specifies the related fifo
out	<i>RxStatusPtr</i>	Informs the caller whether a frame was received (ETH_RECEIVED or ETH_NOT_RECEIVED) and whether more frames are available in the queue (ETH_RECEIVED or ETH_RECEIVED_MORE_DATA_AVAILABLE). implements Eth_Receive_Activity

7.1.3.19 Eth_TxConfirmation()

```
void Eth_TxConfirmation (
    uint8 CtrlIdx )
```

Triggers frame transmission confirmations.

Parameters

in	<i>CtrlIdx</i>	Index of the controller which shall be checked whether any frame transmission has finished. The index is valid within the context of the Ethernet Driver only.
----	----------------	--

Module Documentation

All transmit buffers are checked and upper layers are informed about successfully transmitted frames. Buffers containing transmitted frames are unlocked after the confirmation. implements Eth_TxConfirmation_Activity

7.1.3.20 Eth_GetVersionInfo()

```
void Eth_GetVersionInfo (
    Std_VersionInfoType * VersionInfoPtr )
```

Returns the version information of this module.

Parameters

out	<i>VersionInfoPtr</i>	Pointer where to store the version information of this particular module instance. implements Eth_GetVersionInfo_Activity
-----	-----------------------	--

7.2 GMAC Driver

7.2.1 Detailed Description

7.2.1.1 Platform-specific prerequisites

It is the user's responsibility to configure `SRC_GMAC_0_CTRL_STS[PHY_MODE]` and `SRC_GMAC_0_CTRL_STS[PHY_INTF_SEL]` for GMAC_0 and `SRC_1_GMAC_1_CTRL_REG[PHY_MODE]` and `SRC_1_GMAC_0_CTRL_REG[PHY_INTF_SEL]` for GMAC_1 (just for S32R45) to the desired PHY interface. It shall match the interface expected by the PHY and the MAC.

7.2.1.2 How to use the GMAC driver in your application

The GMAC driver does not handle clock setup or any kind of pin configuration. This is handled by the **Clock Driver** and **Pin Driver** modules, respectively. The driver assumes that the correct clock and pins configurations have been made, so it is the user's responsibility to set up clocking and pin configurations correctly.

In order to use the GMAC driver in your application, the `Gmac_Ip_Init()` function should be called prior to using the rest of the API. The parameters of this function specify:

- The GMAC instance to be initialized
- A structure which will hold the internal state of the driver
- A structure specifying the configuration of the GMAC module
- Two arrays of structures specifying the configuration of the buffers (one structure for each rings - an array for Rx and one for Tx)
- The MAC address to be configured for the module

The configuration of the module is specified through the `Gmac_Ip_ConfigType` structure and contains:

- MII-related configurations (mode, speed, duplex)
- Configuration options for the receive and transmit path
- The maximum frame length
- The MAC interrupt sources which should be enabled
- A callback function to be invoked on events
- The number of Rx and Tx rings used by the driver
- MAC address filtering options

The buffers configuration is specified through the `Gmac_Ip_TxRingConfigType` and `Gmac_Ip_RxRingConfigType` structures and contains:

- The size of the ring

- Pointers to the beginning of the buffer descriptor rings
- Pointers to the beginning of the memory area where the buffer data resides
- Channel interrupt sources
- A callback function to be invoked on per channel events
- The DMA burst length
- A list of VLAN priorities
- The queue operation mode - only for Tx rings
- The scheduling weight/quantum for DCB queues - only for Tx rings
- Idle and send slope credits for AVB queues - only for Tx rings
- High and low credit limits for AVB queues - only for Tx rings

All untagged packets will be received by ring 0 (zero). VLAN tagged packets are routed depending on the VLAN Tag Priority field and the way these priorities were assigned to the rings in the configuration.

In order to de-initialize the driver, the [Gmac_Ip_Deinit\(\)](#) function shall be used. This function will disable the GMAC interrupts and the module, so calling other GMAC driver functions after de-initializing the driver will have undefined behavior. In order to use the driver again, [Gmac_Ip_Init\(\)](#) should be called.

7.2.1.3 How to enable zero-copy optimization

The buffers allocation is controlled by the following configuration parameters:

- EthCtrlAllocateTxDataBuffers
- EthCtrlAllocateRxDataBuffers

Setting them both to FALSE will prevent the driver from using internal buffers.

However, it is important to keep in mind that the programming sequences will also be slightly affected:

- EthAllocateTxDataBuffers = TRUE
 - Call sequence for each frame: `Gmac_Ip_GetTxBuff -> Gmac_Ip_SendFrame`
 - `Gmac_Ip_SendFrame` must be called with the same buffer returned by `Gmac_Ip_GetTxBuff`
- EthAllocateTxDataBuffers = FALSE
 - Call sequence for each frame: `Gmac_Ip_SendFrame`
 - `Gmac_Ip_SendFrame` can be called with any buffer provided by the application
- EthCtrlAllocateRxDataBuffers = TRUE
 - Call sequence for each frame: `Gmac_Ip_ReadFrame -> Gmac_Ip_ProvideRxBuff`

- Gmac_Ip_ProvideRxBuff must be called with the same buffer returned by Gmac_Ip_ReadFrame
- EthCtrlAllocateRxDataBuffers = FALSE
 1. Application has an RX buffer pool
 - After controller initialization: Call Gmac_Ip_ProvideRxBuff for each buffer
 - Call sequence for each frame: Gmac_Ip_ReadFrame -> Gmac_Ip_ProvideRxBuff
 - Gmac_Ip_ProvideRxBuff can be called with either:
 - * The same buffer returned by Gmac_Ip_ReadFrame if the application wants to reuse the buffer
 - * A buffer different than the one returned by Gmac_Ip_ReadFrame if the application doesn't want to reuse the buffer
 2. Application doesn't have an RX buffer pool and wants to provide RX buffers on demand
 - After controller initialization: Nothing to be done
 - Call sequence for each frame: Gmac_Ip_ProvideRxBuff -> Gmac_Ip_ReadFrame

7.2.1.4 How to calculate size of ethernet buffer areas

The contents of this section are relevant only when EthCtrlAllocateTxDataBuffers = TRUE and/or EthCtrlAllocateRxDataBuffers = TRUE.

Ethernet buffer areas is placed in non-cacheable memory for each queue in each controller with order:

- Rx descriptor buffer areas: number of buffers configured(by EthCtrlConfigIngressFifoBufTotal) * size of each buffer (Gmac_Ip_BufferDescriptorType).
- Rx data buffer areas: number of buffers configured(by EthCtrlConfigIngressFifoBufTotal) * size of each buffer (EthCtrlConfigIngressFifoBufLenByte).
- Tx descriptor buffer areas: number of buffers configured(by EthCtrlConfigEgressFifoBufTotal) * size of each buffer (Gmac_Ip_BufferDescriptorType).
- Tx data buffer areas: number of buffers configured(by EthCtrlConfigEgressFifoBufTotal) * size of each buffer (EthCtrlConfigEgressFifoBufLenByte).

All of above memory areas need to align with start address by FEATURE_GMAC_BUFFDESCR_ALIGNMEN←T_BYTES(8U).

This configuration will be placed in loop with multiple queues and multiple controllers.

7.2.1.4.1 In code: - For Reception buffer descriptors for Rx Ring [!"num:i(\$RingIdx)"!] :

```
VAR_ALIGN(Gmac_Ip_BufferDescriptorType GMAC_[!"num:i($CtrlIdx)"!]"_RxRing_[!"num:i($RingIdx)"!]"_DescBuffer[GMA
FEATURE_GMAC_BUFFDESCR_ALIGNMENT_BYTES)
```

- For Reception data buffers for Rx Ring [!"num:i(\$RingIdx)"!]

```
VAR_ALIGN(uint8 GMAC_[!"num:i($CtrlIdx)"!]"_RxRing_[!"num:i($RingIdx)"!]"_DataBuffer[GMAC_[!"num:i($CtrlIdx)"!]"_M
* GMAC_[!"num:i($CtrlIdx)"!]"_MAX_RXBUFFLEN_SUPPORTED], FEATURE_GMAC_BUFFDATA_ALIGNMENT_BY
```

- For Transmission buffer descriptors for Tx Ring [!"num:i(\$RingIdx)"!]

```
VAR_ALIGN(Gmac_Ip_BufferDescriptorType GMAC_[!"num:i($CtrlIdx)"!]"_TxRing_[!"num:i($RingIdx)"!]"_DescBuffer[GMA
FEATURE_GMAC_BUFFDESCR_ALIGNMENT_BYTES)
```

- For Transmission data buffers for Tx Ring [!"num:i(\$RingIdx)"!]

```
VAR_ALIGN(uint8 GMAC_[!"num:i($CtrlIdx)"!]"_TxRing_[!"num:i($RingIdx)"!]"_DataBuffer[GMAC_[!"num:i($CtrlIdx)"!]"_M
* GMAC_[!"num:i($CtrlIdx)"!]"_MAX_TXBUFFLEN_SUPPORTED], FEATURE_GMAC_BUFFDATA_ALIGNMENT_BY
```

7.2.1.5 How to set up MAC address filters

The Address Filtering Module of the MAC checks the source address and destination address fields of each incoming packet. By default, the module only receives the frames having the destination address equal to the MAC address with which the GMAC module has been initialized or the broadcast address.

The module has 1 additional 48-bit perfect address filter with masks for each byte, which can be configured for either source address or destination address filtering. This filter can be set-up by calling the [Gmac_Ip_SetAddrPerfectFilter\(\)](#) function.

Multicast and unicast destination addresses can be added to the hash filter by calling the [Gmac_Ip_AddDstAddrToHashFilter\(\)](#) function and removed by calling the [Gmac_Ip_RemoveDstAddrFromHashFilter\(\)](#) function.

The choice between perfect and hash filters can be done by calling the [Gmac_Ip_SetUnicastHashFilter\(\)](#), [Gmac_Ip_SetMulticastHashFilter\(\)](#) and [Gmac_Ip_SetHashOrPerfectFilter\(\)](#) functions.

For multicast and broadcast frames, the filters can be bypassed by calling the [Gmac_Ip_SetBroadcastForwardAll\(\)](#) and [Gmac_Ip_SetMulticastForwardAll\(\)](#) functions.

7.2.1.6 How to set up VLAN control and filters

Processing of IEEE802.1Q VLAN tags in both transmitted and received packets can be enabled and configured by calling the [Gmac_Ip_EnableVlan\(\)](#) function. This function takes a parameter of type [Gmac_Ip_VlanConfigType](#) which encapsulates the following properties:

- Enables processing of up to two VLAN tags in transmitted and received packets (if present)
- Enables the transmitter and receiver to consider the S-VLAN packets (Type = 0x88A8) as valid VLAN tagged packets
- Selects if outer VLAN tag and type are deleted, inserted or replaced in transmitted packets

- Selects if inner VLAN tag and type are deleted, inserted or replaced in transmitted packets; effective only if double VLAN processing is enabled
- Selects if outer VLAN tag is stripped in received packets
- Selects if inner VLAN tag is stripped in received packets

The user has the option to configure both outer VLAN tag and type for an active transmission ring, to be automatically inserted or replaced for each packet, by calling the [Gmac_Ip_SetTxOuterVlanTagForInsertion\(\)](#) and [Gmac_Ip_SetTxOuterVlanTagForReplacement](#) functions. Additionally, both inner VLAN tag and type can be configured for all transmission rings, by calling the [Gmac_Ip_SetTxInnerVlanTag\(\)](#) function.

By default, all VLAN tagged packets are accepted. The Rx filter can be enabled and configured by calling the [Gmac_Ip_SetVlanTagRxFilter\(\)](#) function. The matching options that can be configured are the following:

- Enable matching on the inner VLAN tag (if present); by default, the outer VLAN tag is compared
- Enable matching for S-VLAN (Type = 0x88A8) packets; by default, matching is performed for C-VLAN (Type = 0x8100) packets
- Disable matching for the VLAN type (only the VLAN tag is compared); by default, the VLAN type is compared
- Enable VLAN tag inverse filter match
- Enable 12-bit VLAN identifier match; by default, the complete 16-bit VLAN tag is compared

The available perfect filters (see the `FEATURE_GMAC_VLAN_RX_FILTERS_NUM` define for the the specific maximum number of filters configurable on each platform) can be set, read and cleared by the means of the following functions: [Gmac_Ip_AddVlanTagToRxFilter\(\)](#), [Gmac_Ip_ReadVlanTagRxFilter\(\)](#), [Gmac_Ip_ClearVlanTagRxFilter\(\)](#). These filters are used for perfect matching and the above described matching criteria are applied to all of them, based on the user configuration. Optionally, a hash table can be used for group matching of 16-bit outer VLAN tags, irrespective of the VLAN type or other criteria. VLAN tags can be added or removed from to/from the hash table by calling the [Gmac_Ip_AddVlanTagToHashTable\(\)](#) or [Gmac_Ip_RemoveVlanTagFromHashTable\(\)](#) functions.

When the Rx filters are enabled, the packets with unmatching VLAN tags will be rejected.

7.2.1.7 How to set up the System Time and enable timestamping for packets

For the timestamp to be captured for Tx and Rx packets, the System Time must be first initialized by calling the [Gmac_Ip_InitSysTime\(\)](#) function. The timestamp is available as extended information in [Gmac_Ip_TxInfoType](#) and [Gmac_Ip_RxInfoType](#) structures, passed as parameters to [Gmac_Ip_GetTransmitStatus\(\)](#) and [Gmac_Ip_ReadFrame\(\)](#).

Please note that the clock source for the GMAC System Time is not handled by the GMAC driver and the user must ensure that it is correctly enabled prior to initializing the System Time.

7.2.1.8 Examples:

7.2.1.8.1 Initializing the module: [Gmac_Ip_Init](#)(INST_GMAC_0, &Gmac_0_ConfigPB_<functionalGroupName>);

```
/* ... */
```

```
Gmac_Ip_Deinit(INST_GMAC_0);
```

7.2.1.8.2 Sending a frame (EthCtrlAllocateTxDataBuffers = TRUE): `Gmac_Ip_BufferType` buffer;

`Gmac_Ip_TxOptionsType` options = {FALSE, GMAC_CRC_AND_PAD_INSERTION, GMAC_CHECKSUM_INSERTION

buffer.length = 8;

if (GMAC_STATUS_SUCCESS == `Gmac_Ip_GetTxBuff`(INST_GMAC_0, 0, &buffer, NULL_PTR))

{

for (uint32 i = 0; i < 8; ++i)

{

buffer.data[i] = i;

}

`Gmac_Ip_SendFrame`(INST_GMAC_0, 0, &buffer, &options);

}

7.2.1.8.3 Sending a frame (EthCtrlAllocateTxDataBuffers = FALSE): `Gmac_Ip_BufferType` buffer;

`Gmac_Ip_TxOptionsType` options = {FALSE, GMAC_CRC_AND_PAD_INSERTION, GMAC_CHECKSUM_INSERTION

uint8_t data[8] = {0, 1, 2, 3, 4, 5, 6, 7};

buffer.data = data;

buffer.length = 8;

`Gmac_Ip_SendFrame`(INST_GMAC_0, 0, &buffer, &options);

7.2.1.8.4 Receiving a frame - polling method (EthCtrlEnableRxInterrupt = FALSE): `Gmac_Ip_BufferType` buffer;

`Gmac_Ip_RxInfoType` info;

for (;;)

{

if (status == `Gmac_Ip_ReadFrame`(INST_GMAC_0, 0, &buffer, &info))

{

/* Process buffer */

/* buffer is no longer needed, provide it to the driver in order to be used by the reception mechanism */

`Gmac_Ip_ProvideRxBuff`(INST_GMAC_0, 0, &buffer);

}

}

7.2.1.8.5 Receiving a frame - interrupt method (EthCtrlEnableRxInterrupt = TRUE): void rx_callback(uint8 instance, uint8 channel)

```
{
    Gmac_Ip_BufferType buffer;

    if (status == Gmac_Ip_ReadFrame(INST_GMAC_0, 0, &buffer, NULL_PTR))
    {
        /* Process buffer */

        /* buffer is no longer needed, provide it to the driver in order to be used by the reception mechanism */
        Gmac_Ip_ProvideRxBuff(INST_GMAC_0, 0, &buffer);
    }
}

int main(void)
{
    /* ... */

    Gmac_Ip_Init(INST_GMAC_0, &Gmac_0_ConfigPB_<functionalGroupName>);

    /* ... */
}
```

7.2.1.8.6 Enabling and configuring VLAN processing and filtering: Gmac_Ip_VlanConfigType gmac_0_vlanConfig = {

```
/* .enDoubleVlan = */ TRUE,

/* .enSvlan = */ FALSE,

/* .outerVlanIns = */ GMAC_VLAN_TAG_INSERTION,

/* .innerVlanIns = */ GMAC_VLAN_TAG_REPLACEMENT,

/* .outerVlanStrip = */ GMAC_VLAN_TAG_DO_NOT_STRIP,

/* .innerVlanStrip = */ GMAC_VLAN_TAG_STRIP_IF_FILTER_FAIL,

};

Gmac_Ip_VlanRxFilterType gmac_0_vlanRxFilterConfig = {

/* .enInnerVlanMatch = */ FALSE,
```



```
/* .enSvlanMatch = */ FALSE,

/* .disVlanTypeMatch = */ FALSE,

/* .enInverseMatch = */ FALSE,

/* .en12bitMatch = */ TRUE,

};

/* Enable VLAN tag processing. */

Gmac_Ip_EnableVlan(INST_GMAC_0, &gmac_0_vlanConfig);

/* Set the outer VLAN tag to inserted for each transmitted packet (a different tag is inserted
depending on the used transmission ring). */

Gmac_Ip_SetTxOuterVlanTagForInsertion(INST_GMAC_0, 0U, GMAC_VLAN_TYPE_C_VLAN, 0x1U);
Gmac_Ip_SetTxOuterVlanTagForInsertion(INST_GMAC_0, 1U, GMAC_VLAN_TYPE_C_VLAN, 0x2U);
Gmac_Ip_SetTxOuterVlanTagForInsertion(INST_GMAC_0, 2U, GMAC_VLAN_TYPE_C_VLAN, 0x3U);

/* Set the inner VLAN tag to be replaced for each transmitted packet. */

Gmac_Ip_SetTxOuterVlanTagForReplacement(INST_GMAC_0, GMAC_VLAN_TYPE_S_VLAN, 0x03U);

/* Enable VLAN tag Rx filters. */

Gmac_Ip_SetVlanTagRxFilter(INST_GMAC_0, TRUE, &gmac_0_vlanRxFilterConfig);

/* Add different VLAN tags to the perfect filters. */

Gmac_Ip_AddVlanTagToRxFilter(INST_GMAC_0, 0U, 0x6003U);
Gmac_Ip_AddVlanTagToRxFilter(INST_GMAC_0, 1U, 0x6001U);
Gmac_Ip_AddVlanTagToRxFilter(INST_GMAC_0, 2U, 0x2U);

/* Add VLAN tags to the hash table for group filtering. */

Gmac_Ip_AddVlanTagToHashTable(INST_GMAC_0, 0x4U);
Gmac_Ip_AddVlanTagToHashTable(INST_GMAC_0, 0x5U);
```

Data Structures

- struct [Gmac_Ip_TxRingConfigType](#)
Specialization of ring configuration for Tx Rings. [More...](#)
- struct [Gmac_Ip_RxRingConfigType](#)
Specialization of ring configuration for Rx Rings. [More...](#)
- struct [Gmac_Ip_BufferType](#)
Send/Receive buffer information for the user. [More...](#)
- struct [Gmac_Ip_TimestampType](#)
Defines the timestamp. [More...](#)
- struct [Gmac_Ip_SysTimeConfigType](#)
Defines a structure to configure the system time. [More...](#)
- struct [Gmac_Ip_TxOptionsType](#)
Transmit options for a particular frame. [More...](#)
- struct [Gmac_Ip_TxInfoType](#)
Detailed status of a transmit buffer descriptor. [More...](#)
- struct [Gmac_Ip_RxInfoType](#)
Detailed status of a receive buffer descriptor. [More...](#)
- struct [Gmac_Ip_VlanConfigType](#)
Defines the VLAN configuration structure. [More...](#)
- struct [Gmac_Ip_VlanRxFilterType](#)
Defines the VLAN filtering and matching options for received packets. [More...](#)

Macros

- [#define GMAC_TX_ERR_IP_HEADER_ERROR](#)
When set, it indicates that the Checksum Offload engine detected an IP header error.
- [#define GMAC_TX_ERR_JABBER_TIMEOUT](#)
When set, it indicates that the MAC transmitter has experienced a jabber time-out.
- [#define GMAC_TX_ERR_PACKET_FLUSHED](#)
When set, it indicates that the GMAC module flushed the packet because of a software flush command given by the CPU.
- [#define GMAC_TX_ERR_PAYLOAD_CHECKSUM_ERROR](#)
When set, it indicates that the Checksum Offload engine had a failure and did not insert any checksum into the encapsulated TCP, UDP, or ICMP payload.
- [#define GMAC_TX_ERR_LOSS_OF_CARRIER](#)
When set, it indicates that Loss of Carrier occurred during packet transmission. This is valid only for the packets transmitted without collision and when the MAC operates in the half-duplex mode.
- [#define GMAC_TX_ERR_NO_CARRIER](#)
When set, it indicates that the carrier sense signal from the PHY was not asserted during transmission.
- [#define GMAC_TX_ERR_LATE_COLLISION](#)
When set, it indicates that packet transmission was aborted because a collision occurred after the collision window (64 byte times including Preamble in MII mode and 512 byte times including Preamble and Carrier Extension in GMII mode). This bit is not valid if Underflow Error is set.
- [#define GMAC_TX_ERR_EXCESSIVE_COLLISION](#)
When set, it indicates that the transmission was aborted after 16 successive collisions while attempting to transmit the current packet.

- `#define GMAC_TX_ERR_EXCESSIVE_DEFERRAL`
When set, it indicates that the transmission ended because of excessive deferral of over 24,288 bit times (155,680 bits times in 1000 Mbps mode or Jumbo Packet enabled mode) if deferral check was set at init.
- `#define GMAC_TX_ERR_UNDERFLOW_ERROR`
When set, it indicates that the MAC aborted the packet because the data arrived late from the system memory.
- `#define GMAC_RX_ERR_CRC_ERROR`
When this bit is set, it indicates that a CRC Error occurred on the received packet.
- `#define GMAC_RX_ERR_DRIBBLE_ERROR`
When this bit is set, it indicates that the received packet has a non-integer multiple of bytes (odd nibbles). This bit is valid only in the MII Mode.
- `#define GMAC_RX_ERR_RECEIVE_ERROR`
When this bit is set, it indicates that the Receive error signal and Receive data valid signal are asserted during packet reception. This error also includes carrier extension error in the GMII and half-duplex mode.
- `#define GMAC_RX_ERR_WATCHDOG_TIMEOUT`
When this bit is set, it indicates that the Receive Watchdog Timer has expired while receiving the current packet. The current packet is truncated after watchdog timeout.
- `#define GMAC_RX_ERR_OVERFLOW_ERROR`
When this bit is set, it indicates that the received packet is damaged because of buffer overflow in Rx FIFO.
- `#define GMAC_RX_ERR_GIANT_PACKET`
When this bit is set, it indicates that the packet length exceeds the specified maximum Ethernet size of 1518, 1522, or 2000 bytes (9018 or 9022 bytes if jumbo packet enable is set).

Types Reference

- `typedef void(* Gmac_Ip_CallbackType) (const uint8 Instance)`
Callback function invoked when a general event is encountered.
- `typedef void(* Gmac_Ip_ChCallbackType) (const uint8 Instance, const uint8 Channel)`
Callback function invoked when a channel event is encountered.

Enum Reference

- `enum Gmac_Ip_StatusType`
Return values.
- `enum Gmac_Ip_CounterType`
Statistics counters enumeration.
- `enum Gmac_Ip_PowerStateType`
Controller power state indication.
- `enum Gmac_Ip_SpeedType`
Media Independent Interface speed selection.
- `enum Gmac_Ip_MacAddrFilterType`
Source/Destination address filter selection.
- `enum Gmac_Ip_CrcPadControlType`
CRC and Pad insertion control.
- `enum Gmac_Ip_ChecksumInsControlType`
Checksum insertion control.
- `enum Gmac_Ip_PayloadType`

- Type of payload encapsulated in the IP datagram.*

 - enum [Gmac_Ip_VlanType](#)
VLAN type.
 - enum [Gmac_Ip_VlanInsControlType](#)
VLAN tag and type deletion, insertion, or replacement control for transmitted packets.
 - enum [Gmac_Ip_VlanStripControlType](#)
VLAN tag stripping control for received packets.
 - enum [Gmac_Ip_SysTimeCorrOffsetType](#)
System time correction offset.
 - enum [Gmac_Ip_TxSchedAlgoType](#)
Transmitter scheduling algorithm.
 - enum [Gmac_Ip_TxThresholdType](#)
Available Tx threshold values.

Initialization and De-initialization

- [Gmac_Ip_StatusType](#) [Gmac_Ip_Init](#) (uint8 Instance, const [Gmac_CtrlConfigType](#) *Config)
Initializes the GMAC module.
- void [Gmac_Ip_Deinit](#) (uint8 Instance)
Deinitializes the GMAC module.
- [Gmac_Ip_PowerStateType](#) [Gmac_Ip_GetPowerState](#) (uint8 Instance)
Gets the current power state of the GMAC module.
- void [Gmac_Ip_SetPowerState](#) (uint8 Instance, [Gmac_Ip_PowerStateType](#) PowerState)
Sets the power state of the GMAC module.
- void [Gmac_Ip_EnableController](#) (uint8 Instance)
Enables all configured transmit and receive buffers and then enables the controller.
- [Gmac_Ip_StatusType](#) [Gmac_Ip_DisableController](#) (uint8 Instance)
Disables the controller and resets all the configured transmit and receive buffers.
- void [Gmac_Ip_SetSpeed](#) (uint8 Instance, [Gmac_Ip_SpeedType](#) Speed)
Sets the speed of the MII interface.
- [Gmac_Ip_StatusType](#) [Gmac_Ip_GetTxBuff](#) (uint8 Instance, uint8 Ring, [Gmac_Ip_BufferType](#) *Buff, uint16 *BuffId)
Provides a transmit buffer to be used by the application for transmission.
- [Gmac_Ip_StatusType](#) [Gmac_Ip_SendFrame](#) (uint8 Instance, uint8 Ring, const [Gmac_Ip_BufferType](#) *Buff, const [Gmac_Ip_TxOptionsType](#) *Options)
Sends an Ethernet frame.
- [Gmac_Ip_StatusType](#) [Gmac_Ip_ReadFrame](#) (uint8 Instance, uint8 Ring, [Gmac_Ip_BufferType](#) *Buff, [Gmac_Ip_RxInfoType](#) *Info)
Reads a received Ethernet frame.
- void [Gmac_Ip_ProvideRxBuff](#) (uint8 Instance, uint8 Ring, const [Gmac_Ip_BufferType](#) *Buff)
Provides a receive buffer to be used by the driver for reception.
- boolean [Gmac_Ip_IsFrameAvailable](#) (uint8 Instance, uint8 Ring)
Checks if there are more frames available in the given queue.
- [Gmac_Ip_StatusType](#) [Gmac_Ip_GetTransmitStatus](#) (uint8 Instance, uint8 Ring, const [Gmac_Ip_BufferType](#) *Buff, [Gmac_Ip_TxInfoType](#) *Info)
Checks if the transmission of a buffer is complete.

- uint32 [Gmac_Ip_GetCounter](#) (uint8 Instance, [Gmac_Ip_CounterType](#) Counter)
Gets statistics from the specified counter.
- void [Gmac_Ip_EnableMDIO](#) (uint8 Instance, boolean MiiPreambleDisabled, uint32 ModuleClk)
Enables the MDIO interface.
- [Gmac_Ip_StatusType](#) [Gmac_Ip_MDIORead](#) (uint8 Instance, uint8 PhyAddr, uint8 PhyReg, uint16 *Data, uint32 TimeoutMs)
Reads the selected register of the PHY.
- [Gmac_Ip_StatusType](#) [Gmac_Ip_MDIOWrite](#) (uint8 Instance, uint8 PhyAddr, uint8 PhyReg, uint16 Data, uint32 TimeoutMs)
Writes the selected register of the PHY.
- [Gmac_Ip_StatusType](#) [Gmac_Ip_MDIOReadMMD](#) (uint8 Instance, uint8 PhyAddr, uint8 Mmd, uint16 PhyReg, uint16 *Data, uint32 TimeoutMs)
Reads a register of the specified MMD in a PHY device.
- [Gmac_Ip_StatusType](#) [Gmac_Ip_MDIOWriteMMD](#) (uint8 Instance, uint8 PhyAddr, uint8 Mmd, uint16 PhyReg, uint16 Data, uint32 TimeoutMs)
Writes a register of the specified MMD in a PHY device.
- void [Gmac_Ip_SetMacAddr](#) (uint8 Instance, const uint8 *MacAddr)
Configures the physical address of the MAC.
- void [Gmac_Ip_GetMacAddr](#) (uint8 Instance, uint8 *MacAddr)
Gets the physical address of the MAC.
- [Gmac_Ip_StatusType](#) [Gmac_Ip_TxTimeAwareShaperInit](#) (uint8 Instance, const [Gmac_CtrlConfigType](#) *Config)
Initialize time aware shaper.
- uint32 [Gmac_Ip_GetInterruptFlags](#) (uint8 Instance)
Gets a mask of the common interrupt events which occurred.
- uint32 [Gmac_Ip_GetChInterruptFlags](#) (uint8 Instance, uint8 Channel)
Gets a mask of the interrupt events associated to a channel which occurred.
- void [Gmac_Ip_SetBroadcastForwardAll](#) (uint8 Instance, boolean Enable)
Enables/Disables forwarding of the broadcast traffic.
- void [Gmac_Ip_SetMulticastForwardAll](#) (uint8 Instance, boolean Enable)
Enables/Disables forwarding of the multicast traffic, irrespective of the destination MAC address.
- void [Gmac_Ip_SetUnicastHashFilter](#) (uint8 Instance, boolean Enable)
Enables/Disables hash filtering for unicast traffic.
- void [Gmac_Ip_SetMulticastHashFilter](#) (uint8 Instance, boolean Enable)
Enables/Disables hash filtering for multicast traffic.
- void [Gmac_Ip_SetHashOrPerfectFilter](#) (uint8 Instance, boolean Enable)
Enables/Disables either hash or perfect filters. If the destination address of an incoming packet passes either the perfect filter or the hash filter, the packet is forwarded to the application.
- void [Gmac_Ip_AddDstAddrToHashFilter](#) (uint8 Instance, const uint8 *MacAddr)
Adds a hardware address to the hash filter. The destination address of an incoming packet is passed through CRC logic and then compared to the entries in the hash table.
- void [Gmac_Ip_RemoveDstAddrFromHashFilter](#) (uint8 Instance, const uint8 *MacAddr)
Removes a hardware address from the hash filter. The destination address of an incoming packet is passed through CRC logic and then compared to the entries in the hash table.
- void [Gmac_Ip_SetAddrPerfectFilter](#) (uint8 Instance, const uint8 *MacAddr, const uint8 Mask, const [Gmac_Ip_MacAddrFilterType](#) AddrType, boolean Enable)
Sets the second destination/source address perfect filter.
- void [Gmac_Ip_SetArpOffloading](#) (uint8 Instance, const uint8 *DstAddr, boolean Enable)

- Enables/Disables ARP offloading.*

 - void [Gmac_Ip_EnableVlan](#) (uint8 Instance, const [Gmac_Ip_VlanConfigType](#) *VlanConfig)

Enable and set VLAN control for transmitted and received packets.

 - [Gmac_Ip_StatusType](#) [Gmac_Ip_SetTxOuterVlanTagForInsertion](#) (uint8 Instance, uint8 Ring, [Gmac_Ip_VlanType](#) VlanType, uint16 VlanTag)

Sets outer VLAN type and tag to be inserted by a particular transmission ring.

 - [Gmac_Ip_StatusType](#) [Gmac_Ip_SetTxOuterVlanTagForReplacement](#) (uint8 Instance, [Gmac_Ip_VlanType](#) VlanType, uint16 VlanTag)

Sets outer VLAN tag to be replaced for all transmission rings.

 - void [Gmac_Ip_SetTxInnerVlanTag](#) (uint8 Instance, [Gmac_Ip_VlanType](#) VlanType, uint16 VlanTag)

Sets inner VLAN type and tag to be inserted/replaced on transmission.

 - void [Gmac_Ip_SetVlanTagRxFilter](#) (uint8 Instance, boolean Enable, const [Gmac_Ip_VlanRxFilterType](#) *RxFilter)

Enables/disables and configures the Rx filters for VLAN tag.

 - [Gmac_Ip_StatusType](#) [Gmac_Ip_AddVlanTagToRxFilter](#) (uint8 Instance, uint8 FilterIdx, uint16 VlanTag)

Adds a VLAN Tag to the Rx filter.

 - [Gmac_Ip_StatusType](#) [Gmac_Ip_ReadVlanTagRxFilter](#) (uint8 Instance, uint8 FilterIdx, uint16 *VlanTag)

Read a VLAN Tag Rx filter.

 - [Gmac_Ip_StatusType](#) [Gmac_Ip_ClearVlanTagRxFilter](#) (uint8 Instance, uint8 FilterIdx)

Clear a VLAN Tag Rx filter.

 - void [Gmac_Ip_AddVlanTagToHashTable](#) (uint8 Instance, uint16 VlanTag)

Adds a VLAN Tag to the Hash Table filter.

 - void [Gmac_Ip_RemoveVlanTagFromHashTable](#) (uint8 Instance, uint16 VlanTag)

Removes a VLAN Tag from the hash table filter.

 - void [Gmac_Ip_InitSysTime](#) (uint8 Instance, const [Gmac_Ip_SysTimeConfigType](#) *SysTimeConfig)

Initialize system time.

 - [Gmac_Ip_StatusType](#) [Gmac_Ip_SetSysTimeCorr](#) (uint8 Instance, [Gmac_Ip_SysTimeCorrOffsetType](#) Offset, uint32 SecondsUpdate, uint32 NanoSecondsUpdate)

Set system time correction.

 - [Gmac_Ip_StatusType](#) [Gmac_Ip_SetRateRatioCorr](#) (uint8 Instance, float64 RateRatio)

Set system time correction.

 - void [Gmac_Ip_GetSysTime](#) (uint8 Instance, [Gmac_Ip_TimestampType](#) *Timestamp)

Gets the current system time.

 - void [Gmac_Ip_SetTxSchedAlgo](#) (uint8 Instance, [Gmac_Ip_TxSchedAlgoType](#) SchedAlgo)

Sets the transmitter scheduling algorithm.

 - void [Gmac_Ip_SetTxRingWeight](#) (uint8 Instance, uint8 Ring, uint32 Weight)

Sets the weight (in WRR scheduling algorithm) for a particular Tx ring.

 - void [Gmac_Ip_EnableTxStoreAndForward](#) (uint8 Instance, uint8 Ring)

Enables the store and forward feature on the transmit path.

 - void [Gmac_Ip_SetTxThreshold](#) (uint8 Instance, uint8 Ring, [Gmac_Ip_TxThresholdType](#) ThresholdValue)

Sets transmit threshold levels.

 - #define **ETH_STOP_SEC_CODE**

7.2.2 Data Structure Documentation

7.2.2.1 struct [Gmac_Ip_TxRingConfigType](#)

Specialization of ring configuration for Tx Rings.

Definition at line 569 of file [Gmac_Ip_Types.h](#).

Data Fields

	Type	Name	Description
	uint32	Weight	Scheduling weight/quantum for DCB queues
	uint32	IdleSlopeCredit	Idle slope credit for AVB queues
	uint32	SendSlopeCredit	Send slope credit for AVB queues
	uint32	HiCredit	High credit limit for AVB queues
	sint32	LoCredit	Low credit limit for AVB queues
Gmac_Ip_BufferDescriptorType *		RingDesc	Buffer descriptor ring start address.
Gmac_Ip_ChCallbackType		Callback	Callback function for current channel.
uint8 *		Buffer	Buffer data pool start address
uint32		Interrupts	Channel interrupt sources. A logical OR of "Gmac_Ip_ChInterruptType".
uint16		BufferLen	Length of each individual buffer in a pool
uint16		RingSize	Buffer descriptors number.
uint8		PriorityMask	A logical OR of "Gmac_Ip_VlanPriorityType".
uint8		DmaBurstLength	Maximum number of beats in one DMA block data transfer.
Gmac_Ip_OperationModeType		QueueOpMode	Queue operation mode

7.2.2.2 struct Gmac_Ip_RxRingConfigType

Specialization of ring configuration for Rx Rings.

Definition at line 592 of file Gmac_Ip_Types.h.

Data Fields

- Gmac_Ip_BufferDescriptorType * [RingDesc](#)
- [Gmac_Ip_ChCallbackType](#) [Callback](#)
- uint8 * [Buffer](#)
- uint32 [Interrupts](#)
- uint16 [BufferLen](#)
- uint16 [RingSize](#)
- uint8 [PriorityMask](#)
- uint8 [DmaBurstLength](#)

7.2.2.2.1 Field Documentation

7.2.2.2.1.1 RingDesc Gmac_Ip_BufferDescriptorType* RingDesc

Buffer descriptor ring start address.

Definition at line 594 of file Gmac_Ip_Types.h.

7.2.2.2.1.2 Callback `Gmac_Ip_ChCallbackType` Callback

Callback function for current channel.

Definition at line 595 of file `Gmac_Ip_Types.h`.

7.2.2.2.1.3 Buffer `uint8* Buffer`

Buffer data pool start address

Definition at line 596 of file `Gmac_Ip_Types.h`.

7.2.2.2.1.4 Interrupts `uint32 Interrupts`

Channel interrupt sources. A logical OR of "Gmac_Ip_ChInterruptType".

Definition at line 597 of file `Gmac_Ip_Types.h`.

7.2.2.2.1.5 BufferLen `uint16 BufferLen`

Length of each individual buffer in a pool

Definition at line 598 of file `Gmac_Ip_Types.h`.

7.2.2.2.1.6 RingSize `uint16 RingSize`

Buffer descriptors number.

Definition at line 599 of file `Gmac_Ip_Types.h`.

7.2.2.2.1.7 PriorityMask `uint8 PriorityMask`

A logical OR of "Gmac_Ip_VlanPriorityType".

Definition at line 600 of file `Gmac_Ip_Types.h`.

7.2.2.2.1.8 DmaBurstLength `uint8 DmaBurstLength`

Maximum number of beats in one DMA block data transfer.

Definition at line 601 of file Gmac_Ip_Types.h.

7.2.2.3 struct Gmac_Ip_BufferType

Send/Receive buffer information for the user.

Definition at line 609 of file Gmac_Ip_Types.h.

Data Fields

- `uint8 * Data`
- `uint16 Length`

7.2.2.3.1 Field Documentation

7.2.2.3.1.1 Data `uint8* Data`

Pointer to the data buffer

Definition at line 611 of file Gmac_Ip_Types.h.

7.2.2.3.1.2 Length `uint16 Length`

Length of the data buffer

Definition at line 612 of file Gmac_Ip_Types.h.

7.2.2.4 struct Gmac_Ip_TimestampType

Defines the timestamp.

Definition at line 619 of file Gmac_Ip_Types.h.

Data Fields

- `uint32 nanoseconds`
- `uint32 seconds`
- `uint16 secondsHi`

7.2.2.4.1 Field Documentation

7.2.2.4.1.1 nanoseconds `uint32 nanoseconds`

Nanoseconds part of the time.

Definition at line 621 of file `Gmac_Ip_Types.h`.

7.2.2.4.1.2 seconds `uint32 seconds`

The 32 least significant bits of the 48 bits seconds part of the time.

Definition at line 622 of file `Gmac_Ip_Types.h`.

7.2.2.4.1.3 secondsHi `uint16 secondsHi`

The 16 most significant bits of the 48 bit seconds part of the time.

Definition at line 623 of file `Gmac_Ip_Types.h`.

7.2.2.5 struct `Gmac_Ip_SysTimeConfigType`

Defines a structure to configure the system time.

Definition at line 630 of file `Gmac_Ip_Types.h`.

Data Fields

- `uint8 SubSecondInc`
- `uint8 SubNanoSecondsInc`
- `Gmac_Ip_TimestampType * InitialTimestamp`

7.2.2.5.1 Field Documentation

7.2.2.5.1.1 SubSecondInc `uint8 SubSecondInc`

The value in nanoseconds to be accumulated every clock cycle.

Definition at line 632 of file `Gmac_Ip_Types.h`.

7.2.2.5.1.2 SubNanoSecondsInc `uint8 SubNanoSecondsInc`

The value in sub-nanoseconds to be accumulated every clock cycle, represented in nanoseconds multiplied by 2^8 . For example, if the required increment is 5.3ns, then subSecondInc should be 0x05 and subNanosecondsInc should be 0x4C ($0.3 * 2^8$).

Definition at line 633 of file Gmac_Ip_Types.h.

7.2.2.5.1.3 InitialTimestamp `Gmac_Ip_TimestampType* InitialTimestamp`

Nanoseconds part of the time.

Definition at line 636 of file Gmac_Ip_Types.h.

7.2.2.6 struct Gmac_Ip_TxOptionsType

Transmit options for a particular frame.

Definition at line 643 of file Gmac_Ip_Types.h.

Data Fields

- boolean `NoInt`
- `Gmac_Ip_CrcPadControlType CrcPadIns`
- `Gmac_Ip_ChecksumInsControlType ChecksumIns`

7.2.2.6.1 Field Documentation

7.2.2.6.1.1 NoInt `boolean NoInt`

Does not generate interrupt on transmit completion.

Definition at line 645 of file Gmac_Ip_Types.h.

7.2.2.6.1.2 CrcPadIns `Gmac_Ip_CrcPadControlType CrcPadIns`

Controls the CRC and pad insertion for Tx packet.

Definition at line 646 of file Gmac_Ip_Types.h.

7.2.2.6.1.3 ChecksumIns `Gmac_Ip_ChecksumInsControlType` ChecksumIns

Controls the IP checksum insertion for Tx packet.

Definition at line 647 of file `Gmac_Ip_Types.h`.

7.2.2.7 struct `Gmac_Ip_TxInfoType`

Detailed status of a transmit buffer descriptor.

Definition at line 654 of file `Gmac_Ip_Types.h`.

Data Fields

- uint32 `ErrMask`
- `Gmac_Ip_TimestampType` Timestamp

7.2.2.7.1 Field Documentation

7.2.2.7.1.1 ErrMask `uint32` ErrMask

Mask of `GMAC_TX_ERR_*`, representing the errors that occurred.

Definition at line 656 of file `Gmac_Ip_Types.h`.

7.2.2.7.1.2 Timestamp `Gmac_Ip_TimestampType` Timestamp

The timestamp of the transmitted packet.

Definition at line 657 of file `Gmac_Ip_Types.h`.

7.2.2.8 struct `Gmac_Ip_RxInfoType`

Detailed status of a receive buffer descriptor.

Definition at line 706 of file `Gmac_Ip_Types.h`.

Data Fields

- uint32 [ErrMask](#)
- uint16 [PktLen](#)
- [Gmac_Ip_PayloadType](#) PayloadType
- boolean [Ipv6](#)
- boolean [Ipv4](#)
- boolean [IpPayloadErr](#)
- boolean [IpHeaderErr](#)
- uint16 [OuterVlanTag](#)
- uint16 [InnerVlanTag](#)
- [Gmac_Ip_TimestampType](#) Timestamp

7.2.2.8.1 Field Documentation

7.2.2.8.1.1 ErrMask `uint32 ErrMask`

Mask of `GMAC_RX_ERR_*`, representing the errors that occurred.

Definition at line 708 of file `Gmac_Ip_Types.h`.

7.2.2.8.1.2 PktLen `uint16 PktLen`

Byte length of the received packet that was transferred to system memory (including CRC).

Definition at line 709 of file `Gmac_Ip_Types.h`.

7.2.2.8.1.3 PayloadType `Gmac_Ip_PayloadType PayloadType`

The type of payload encapsulated in the IP datagram (this field is valid only if the Checksum Offloading feature is enabled when initializing the GMAC module).

Definition at line 710 of file `Gmac_Ip_Types.h`.

7.2.2.8.1.4 Ipv6 `boolean Ipv6`

Indicates that the frame is an IPv6 frame (this field is valid only if the Checksum Offloading feature is enabled when initializing the GMAC module).

Definition at line 712 of file `Gmac_Ip_Types.h`.

7.2.2.8.1.5 Ipv4 `boolean Ipv4`

Indicates that the frame is an IPv4 frame (this field is valid only if the Checksum Offloading feature is enabled when initializing the GMAC module).

Definition at line 714 of file `Gmac_Ip_Types.h`.

7.2.2.8.1.6 IpPayloadErr `boolean IpPayloadErr`

Indicates that an IP payload error has occurred (this field is valid only if the Checksum Offloading feature is enabled when initializing the GMAC module).

Definition at line 716 of file `Gmac_Ip_Types.h`.

7.2.2.8.1.7 IpHeaderErr `boolean IpHeaderErr`

Indicates that an IP header error has occurred (this field is valid only if the Checksum Offloading feature is enabled when initializing the GMAC module).

Definition at line 718 of file `Gmac_Ip_Types.h`.

7.2.2.8.1.8 OuterVlanTag `uint16 OuterVlanTag`

Contains the outer VLAN tag of the received packet.

Definition at line 720 of file `Gmac_Ip_Types.h`.

7.2.2.8.1.9 InnerVlanTag `uint16 InnerVlanTag`

Contains the inner VLAN tag of the received packet. This is valid only when Double VLAN tag processing is enabled.

Definition at line 721 of file `Gmac_Ip_Types.h`.

7.2.2.8.1.10 Timestamp `Gmac_Ip_TimestampType Timestamp`

The timestamp of the received packet.

Definition at line 723 of file `Gmac_Ip_Types.h`.

7.2.2.9 struct Gmac_Ip_VlanConfigType

Defines the VLAN configuration structure.

Definition at line 754 of file Gmac_Ip_Types.h.

Data Fields

- boolean [EnDoubleVlan](#)
- boolean [EnSvlan](#)
- [Gmac_Ip_VlanInsControlType](#) OuterVlanIns
- [Gmac_Ip_VlanInsControlType](#) InnerVlanIns
- [Gmac_Ip_VlanStripControlType](#) OuterVlanStrip
- [Gmac_Ip_VlanStripControlType](#) InnerVlanStrip

7.2.2.9.1 Field Documentation

7.2.2.9.1.1 EnDoubleVlan `boolean EnDoubleVlan`

Enables processing of up to two VLAN tags on Tx and Rx (if present).

Definition at line 756 of file Gmac_Ip_Types.h.

7.2.2.9.1.2 EnSvlan `boolean EnSvlan`

Enables MAC transmitter and receiver to consider the S-VLAN packets (Type = 0x88A8) as valid VLAN tagged packets.

Definition at line 757 of file Gmac_Ip_Types.h.

7.2.2.9.1.3 OuterVlanIns `Gmac_Ip_VlanInsControlType OuterVlanIns`

Selects if outer VLAN tag and type are deleted, inserted or replaced in transmitted packets.

Definition at line 758 of file Gmac_Ip_Types.h.

7.2.2.9.1.4 InnerVlanIns `Gmac_Ip_VlanInsControlType InnerVlanIns`

Selects if inner VLAN tag and type are deleted, inserted or replaced in transmitted packets. Effective only if double VLAN processing is enabled.

Definition at line 759 of file Gmac_Ip_Types.h.

7.2.2.9.1.5 OuterVlanStrip `Gmac_Ip_VlanStripControlType` `OuterVlanStrip`

Selects if outer VLAN tag is stripped in received packets.

Definition at line 761 of file `Gmac_Ip_Types.h`.

7.2.2.9.1.6 InnerVlanStrip `Gmac_Ip_VlanStripControlType` `InnerVlanStrip`

Selects if inner VLAN tag is stripped in received packets.

Definition at line 762 of file `Gmac_Ip_Types.h`.

7.2.2.10 struct Gmac_Ip_VlanRxFilterType

Defines the VLAN filtering and matching options for received packets.

Definition at line 769 of file `Gmac_Ip_Types.h`.

Data Fields

- boolean `EnInnerVlanMatch`
- boolean `EnSvlanMatch`
- boolean `DisVlanTypeMatch`
- boolean `EnInverseMatch`
- boolean `En12bitMatch`

7.2.2.10.1 Field Documentation**7.2.2.10.1.1 EnInnerVlanMatch** `boolean` `EnInnerVlanMatch`

When true, the MAC receiver enables operation on the inner VLAN tag (if present). When false, the MAC receiver enables operation on the outer VLAN tag (if present).

Definition at line 771 of file `Gmac_Ip_Types.h`.

7.2.2.10.1.2 EnSvlanMatch `boolean` `EnSvlanMatch`

When true, the MAC receiver enables filtering or matching for S-VLAN (Type = 0x88A8) packets. When false, the MAC receiver enables filtering or matching for C-VLAN (Type = 0x8100) packets.

Definition at line 773 of file `Gmac_Ip_Types.h`.

7.2.2.10.1.3 DisVlanTypeMatch `boolean DisVlanTypeMatch`

When true, the MAC does not check whether the inner VLAN tag (if inner VLAN match is enabled) is of type S-VLAN or C-VLAN. When false, the MAC filters or matches the inner VLAN tag (if inner VLAN match is enabled) only when VLAN tag type is similar to the one specified by the Enable S-VLAN Match for received Frames.

Definition at line 775 of file Gmac_Ip_Types.h.

7.2.2.10.1.4 EnInverseMatch `boolean EnInverseMatch`

When true, enables VLAN tag inverse filter match.

Definition at line 778 of file Gmac_Ip_Types.h.

7.2.2.10.1.5 En12bitMatch `boolean En12bitMatch`

When true, a 12-bit VLAN identifier is used for comparing and filtering instead of the complete 16-bit VLAN tag. When false, the complete 16-bit VLAN tag is compared.

Definition at line 779 of file Gmac_Ip_Types.h.

7.2.3 Macro Definition Documentation

7.2.3.1 GMAC_TX_ERR_IP_HEADER_ERROR

```
#define GMAC_TX_ERR_IP_HEADER_ERROR
```

When set, it indicates that the Checksum Offload engine detected an IP header error.

Definition at line 661 of file Gmac_Ip_Types.h.

7.2.3.2 GMAC_TX_ERR_JABBER_TIMEOUT

```
#define GMAC_TX_ERR_JABBER_TIMEOUT
```

When set, it indicates that the MAC transmitter has experienced a jabber time-out.

Definition at line 664 of file Gmac_Ip_Types.h.

7.2.3.3 GMAC_TX_ERR_PACKET_FLUSHED

```
#define GMAC_TX_ERR_PACKET_FLUSHED
```

When set, it indicates that the GMAC module flushed the packet because of a software flush command given by the CPU.

Definition at line 668 of file Gmac_Ip_Types.h.

7.2.3.4 GMAC_TX_ERR_PAYLOAD_CHECKSUM_ERROR

```
#define GMAC_TX_ERR_PAYLOAD_CHECKSUM_ERROR
```

When set, it indicates that the Checksum Offload engine had a failure and did not insert any checksum into the encapsulated TCP, UDP, or ICMP payload.

Definition at line 672 of file Gmac_Ip_Types.h.

7.2.3.5 GMAC_TX_ERR_LOSS_OF_CARRIER

```
#define GMAC_TX_ERR_LOSS_OF_CARRIER
```

When set, it indicates that Loss of Carrier occurred during packet transmission. This is valid only for the packets transmitted without collision and when the MAC operates in the half-duplex mode.

Definition at line 677 of file Gmac_Ip_Types.h.

7.2.3.6 GMAC_TX_ERR_NO_CARRIER

```
#define GMAC_TX_ERR_NO_CARRIER
```

When set, it indicates that the carrier sense signal from the PHY was not asserted during transmission.

Definition at line 681 of file Gmac_Ip_Types.h.

7.2.3.7 GMAC_TX_ERR_LATE_COLLISION

```
#define GMAC_TX_ERR_LATE_COLLISION
```

When set, it indicates that packet transmission was aborted because a collision occurred after the collision window (64 byte times including Preamble in MII mode and 512 byte times including Preamble and Carrier Extension in GMII mode). This bit is not valid if Underflow Error is set.

Definition at line 687 of file Gmac_Ip_Types.h.

7.2.3.8 GMAC_TX_ERR_EXCESSIVE_COLLISION

```
#define GMAC_TX_ERR_EXCESSIVE_COLLISION
```

When set, it indicates that the transmission was aborted after 16 successive collisions while attempting to transmit the current packet.

Definition at line 691 of file Gmac_Ip_Types.h.

7.2.3.9 GMAC_TX_ERR_EXCESSIVE_DEFERRAL

```
#define GMAC_TX_ERR_EXCESSIVE_DEFERRAL
```

When set, it indicates that the transmission ended because of excessive deferral of over 24,288 bit times (155,680 bits times in 1000 Mbps mode or Jumbo Packet enabled mode) if deferral check was set at init.

Definition at line 696 of file Gmac_Ip_Types.h.

7.2.3.10 GMAC_TX_ERR_UNDERFLOW_ERROR

```
#define GMAC_TX_ERR_UNDERFLOW_ERROR
```

When set, it indicates that the MAC aborted the packet because the data arrived late from the system memory.

Definition at line 700 of file Gmac_Ip_Types.h.

7.2.3.11 GMAC_RX_ERR_CRC_ERROR

```
#define GMAC_RX_ERR_CRC_ERROR
```

When this bit is set, it indicates that a CRC Error occurred on the received packet.

Definition at line 727 of file Gmac_Ip_Types.h.

7.2.3.12 GMAC_RX_ERR_DRIBBLE_ERROR

```
#define GMAC_RX_ERR_DRIBBLE_ERROR
```

When this bit is set, it indicates that the received packet has a non-integer multiple of bytes (odd nibbles). This bit is valid only in the MII Mode.

Definition at line 731 of file Gmac_Ip_Types.h.

7.2.3.13 GMAC_RX_ERR_RECEIVE_ERROR

```
#define GMAC_RX_ERR_RECEIVE_ERROR
```

When this bit is set, it indicates that the Receive error signal and Receive data valid signal are asserted during packet reception. This error also includes carrier extension error in the GMII and half-duplex mode.

Definition at line 736 of file Gmac_Ip_Types.h.

7.2.3.14 GMAC_RX_ERR_WATCHDOG_TIMEOUT

```
#define GMAC_RX_ERR_WATCHDOG_TIMEOUT
```

When this bit is set, it indicates that the Receive Watchdog Timer has expired while receiving the current packet. The current packet is truncated after watchdog timeout.

Definition at line 740 of file Gmac_Ip_Types.h.

7.2.3.15 GMAC_RX_ERR_OVERFLOW_ERROR

```
#define GMAC_RX_ERR_OVERFLOW_ERROR
```

When this bit is set, it indicates that the received packet is damaged because of buffer overflow in Rx FIFO.

Definition at line 744 of file Gmac_Ip_Types.h.

7.2.3.16 GMAC_RX_ERR_GIANT_PACKET

```
#define GMAC_RX_ERR_GIANT_PACKET
```

When this bit is set, it indicates that the packet length exceeds the specified maximum Ethernet size of 1518, 1522, or 2000 bytes (9018 or 9022 bytes if jumbo packet enable is set).

Definition at line 748 of file Gmac_Ip_Types.h.

7.2.4 Types Reference

7.2.4.1 Gmac_Ip_CallbackType

```
typedef void(* Gmac_Ip_CallbackType) (const uint8 Instance)
```

Callback function invoked when a general event is encountered.

Definition at line 511 of file Gmac_Ip_Types.h.

7.2.4.2 Gmac_Ip_ChCallbackType

```
typedef void(* Gmac_Ip_ChCallbackType) (const uint8 Instance, const uint8 Channel)
```

Callback function invoked when a channel event is encountered.

Definition at line 516 of file Gmac_Ip_Types.h.

7.2.5 Enum Reference

7.2.5.1 Gmac_Ip_StatusType

```
enum Gmac_Ip_StatusType
```

Return values.

Enumerator

GMAC_STATUS_SUCCESS	Generic operation success status
GMAC_STATUS_ERROR	Generic operation failure status
GMAC_STATUS_BUSY	Generic operation busy status
GMAC_STATUS_TIMEOUT	Generic operation timeout status
GMAC_STATUS_UNSUPPORTED	Generic operation unsupported status
GMAC_STATUS_RX_QUEUE_EMPTY	There is no available frame in the receive queue
GMAC_STATUS_TX_QUEUE_FULL	There is no available space for the frame in the transmit queue
GMAC_STATUS_BUFF_NOT_FOUND	The specified buffer was not found in the queue
GMAC_STATUS_TX_BUFF_BUSY	All internal TX buffers are currently in use
GMAC_STATUS_TX_BUFF_OVERFLOW	The requested TX buffer length cannot

Definition at line 100 of file Gmac_Ip_Types.h.

7.2.5.2 Gmac_Ip_CounterType

enum `Gmac_Ip_CounterType`

Statistics counters enumeration.

Definition at line 236 of file Gmac_Ip_Types.h.

7.2.5.3 Gmac_Ip_PowerStateType

enum `Gmac_Ip_PowerStateType`

Controller power state indication.

Definition at line 297 of file Gmac_Ip_Types.h.

7.2.5.4 Gmac_Ip_SpeedType

enum `Gmac_Ip_SpeedType`

Media Independent Interface speed selection.

Enumerator

GMAC_SPEED_1G	Speed 1 Gbps.
GMAC_SPEED_10M	Speed 10 Mbps.
GMAC_SPEED_100M	Speed 100 Mbps.
GMAC_SPEED_2500M	Speed 2500 Mbps.

Definition at line 309 of file Gmac_Ip_Types.h.

7.2.5.5 Gmac_Ip_MacAddrFilterType

```
enum Gmac_Ip_MacAddrFilterType
```

Source/Destination address filter selection.

Enumerator

GMAC_DST_ADDR_FILTER	Destination address filter.
GMAC_SRC_ADDR_FILTER	Source address filter.
GMAC_SRC_ADDR_INV_FILTER	Source address inverse filter.

Definition at line 353 of file Gmac_Ip_Types.h.

7.2.5.6 Gmac_Ip_CrcPadControlType

```
enum Gmac_Ip_CrcPadControlType
```

CRC and Pad insertion control.

Enumerator

GMAC_CRC_AND_PAD_INSERTION	Appends the CRC at the end of the transmitted packet of length greater than or equal to 60 bytes. Appends padding and CRC to a packet with length less than 60 bytes.
GMAC_CRC_INSERTION	Appends the CRC at the end of the transmitted packet but it does not append padding. The application should ensure that the padding bytes are present in the packet being transferred.
GMAC_CRC_INSERTION_DISABLE	Does not append the CRC at the end of the transmitted packet. The application should ensure that the padding and CRC bytes are present in the packet being transferred.
GMAC_CRC_REPLACEMENT	Replaces the last four bytes of the transmitted packet with recalculated CRC bytes. The application should ensure that the padding and CRC bytes are present in the packet being transferred.

Definition at line 364 of file Gmac_Ip_Types.h.

7.2.5.7 Gmac_Ip_ChecksumInsControlType

```
enum Gmac_Ip_ChecksumInsControlType
```

Checksum insertion control.

Enumerator

GMAC_CHECKSUM_INSERTION_DISABLE	Checksum insertion is disabled.
GMAC_CHECKSUM_INSERTION_IP	IP header checksum calculation and insertion are enabled.
GMAC_CHECKSUM_INSERTION_PROTO_N↔ O_PSEUDOH	Protocol checksum calculation and insertion are enabled, but pseudo-header checksum is not calculated in hardware.
GMAC_CHECKSUM_INSERTION_PROTO_PS↔ EUDOH	Protocol checksum calculation and insertion are enabled, and pseudo-header checksum is calculated in hardware.

Definition at line 380 of file Gmac_Ip_Types.h.

7.2.5.8 Gmac_Ip_PayloadType

```
enum Gmac_Ip_PayloadType
```

Type of payload encapsulated in the IP datagram.

Enumerator

GMAC_IP_PAYLOAD_TYPE_UNKNOWN	Unknown type or IP payload not processed.
GMAC_IP_PAYLOAD_TYPE_UDP	UDP packet.
GMAC_IP_PAYLOAD_TYPE_TCP	TCP packet.
GMAC_IP_PAYLOAD_TYPE_ICMP	ICMP packet.

Definition at line 394 of file Gmac_Ip_Types.h.

7.2.5.9 Gmac_Ip_VlanType

```
enum Gmac_Ip_VlanType
```


VLAN type.

Enumerator

GMAC_VLAN_TYPE_C_VLAN	C-VLAN type (0x8100).
GMAC_VLAN_TYPE_S_VLAN	S-VLAN type (0x88A8).

Definition at line 406 of file Gmac_Ip_Types.h.

7.2.5.10 Gmac_Ip_VlanInsControlType

```
enum Gmac_Ip_VlanInsControlType
```

VLAN tag and type deletion, insertion, or replacement control for transmitted packets.

Enumerator

GMAC_VLAN_TAG_NO_CONTROL	The VLAN tag is not altered in the packet to be transmitted.
GMAC_VLAN_TAG_DELETION	The VLAN tag is deleted from the packet to be transmitted.
GMAC_VLAN_TAG_INSERTION	The VLAN tag is inserted in the packet to be transmitted.
GMAC_VLAN_TAG_REPLACEMENT	The VLAN tag is replaced in the packet to be transmitted.

Definition at line 416 of file Gmac_Ip_Types.h.

7.2.5.11 Gmac_Ip_VlanStripControlType

```
enum Gmac_Ip_VlanStripControlType
```

VLAN tag stripping control for received packets.

Enumerator

GMAC_VLAN_TAG_DO_NOT_STRIP	Does not strip the inner VLAN tag in the received packet.
GMAC_VLAN_TAG_STRIP_IF_FILTER_PASS	Strips the inner VLAN tag in the received packet if the VLAN filter passes.
GMAC_VLAN_TAG_STRIP_IF_FILTER_FAIL	Strips the inner VLAN tag in the received packet if the VLAN filter fails.
GMAC_VLAN_TAG_ALWAYS_STRIP	Always strips the inner VLAN tag in the received packet.

Definition at line 428 of file Gmac_Ip_Types.h.

7.2.5.12 Gmac_Ip_SysTimeCorrOffsetType

enum `Gmac_Ip_SysTimeCorrOffsetType`

System time correction offset.

Enumerator

GMAC_SYS_TIME_CORR_POSITIVE	Correction value is added to the system time.
GMAC_SYS_TIME_CORR_NEGATIVE	Correction value is subtracted from the system time.

Definition at line 457 of file `Gmac_Ip_Types.h`.

7.2.5.13 Gmac_Ip_TxSchedAlgoType

enum `Gmac_Ip_TxSchedAlgoType`

Transmitter scheduling algorithm.

Enumerator

GMAC_SCHED_ALGO_WRR	Weighted round robin algorithm.
GMAC_SCHED_ALGO_WFQ	Weighted fair queueing algorithm.
GMAC_SCHED_ALGO_DWRR	Deficit weighted round robin algorithm.
GMAC_SCHED_ALGO_SP	Strict priority algorithm.

Definition at line 467 of file `Gmac_Ip_Types.h`.

7.2.5.14 Gmac_Ip_TxThresholdType

enum `Gmac_Ip_TxThresholdType`

Available Tx threshold values.

Definition at line 478 of file `Gmac_Ip_Types.h`.

7.2.6 Function Reference

7.2.6.1 Gmac_Ip_Init()

```
Gmac_Ip_StatusType Gmac_Ip_Init (
    uint8 Instance,
    const Gmac_CtrlConfigType * Config )
```

Initializes the GMAC module.

This function initializes and enables the GMAC module, configuring receive and transmit control settings, the receive and transmit descriptors rings, and the MAC physical address.

Note: All untagged packets will be received by ring 0 (zero). VLAN tagged packets are routed depending on the VLAN Tag Priority field according to the provided configuration.

Parameters

in	<i>instance</i>	Instance number
in	<i>config</i>	Pointer to the module configuration structure

Return values

<i>GMAC_STATUS_SUCCESS</i>	The initialization was successful.
<i>GMAC_STATUS_TIMEOUT</i>	The DMA subsystem reset could not complete.

7.2.6.2 Gmac_Ip_Deinit()

```
void Gmac_Ip_Deinit (
    uint8 Instance )
```

Deinitializes the GMAC module.

This function disables the interrupts and then disables the GMAC module.

Parameters

in	<i>instance</i>	Instance number
----	-----------------	-----------------

7.2.6.3 Gmac_Ip_GetPowerState()

```
Gmac_Ip_PowerStateType Gmac_Ip_GetPowerState (
    uint8 Instance )
```

Gets the current power state of the GMAC module.

Parameters

in	<i>instance</i>	Instance number
----	-----------------	-----------------

Returns

Gmac_Ip_PowerStateType The power state of the controller

7.2.6.4 Gmac_Ip_SetPowerState()

```
void Gmac_Ip_SetPowerState (
    uint8 Instance,
    Gmac_Ip_PowerStateType PowerState )
```

Sets the power state of the GMAC module.

Parameters

in	<i>instance</i>	Instance number
in	<i>powerState</i>	Power state to transition into

7.2.6.5 Gmac_Ip_EnableController()

```
void Gmac_Ip_EnableController (
    uint8 Instance )
```

Enables all configured transmit and receive buffers and then enables the controller.

Parameters

in	<i>instance</i>	Instance number
----	-----------------	-----------------

7.2.6.6 Gmac_Ip_DisableController()

```
Gmac_Ip_StatusType Gmac_Ip_DisableController (
    uint8 Instance )
```

Disables the controller and resets all the configured transmit and receive buffers.

Warning: This function ignores all pending transmission and reception requests

Parameters

in	<i>instance</i>	Instance number
----	-----------------	-----------------

Return values

<i>GMAC_STATUS_SUCCESS</i>	Tthe controller was successfully disabled.
<i>GMAC_STATUS_TIMEOUT</i>	Tthe underlying MTL queues could not be flushed.

7.2.6.7 Gmac_Ip_SetSpeed()

```
void Gmac_Ip_SetSpeed (
    uint8 Instance,
    Gmac_Ip_SpeedType Speed )
```

Sets the speed of the MII interface.

Parameters

in	<i>instance</i>	Instance number
in	<i>speed</i>	MI speed

7.2.6.8 Gmac_Ip_GetTxBuff()

```
Gmac_Ip_StatusType Gmac_Ip_GetTxBuff (
    uint8 Instance,
    uint8 Ring,
    Gmac_Ip_BufferType * Buff,
    uint16 * BuffId )
```

Provides a transmit buffer to be used by the application for transmission.

This function provides an internal buffer which can further be used by the application to store the transmit data.

Note: The buffer will be marked as locked and won't be released until after a call to Gmac_Ip_GetTransmitStatus for the same buffer returns GMAC_STATUS_SUCCESS.

Important: The driver does not ensure synchronization between different threads trying to get a buffer at the same time. This synchronization shall be implemented by the application.

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Parameters

in	<i>instance</i>	Instance number
in	<i>ring</i>	Ring number
in, out	<i>buff</i>	In: Buffer containing the desired length Out: Buffer containing the granted length or available length in case of overflow.
out	<i>buffId</i>	Index of the buffer (descriptor) within the ring. If this information is not needed, this parameter should be NULL_PTR.

Return values

<i>GMAC_STATUS_SUCCESS</i>	The buffer has been successfully locked.
<i>GMAC_STATUS_TX_BUFF_BUSY</i>	All buffers are currently in use.
<i>GMAC_STATUS_TX_BUFF_OVERFLOW</i>	The requested buffer length cannot be granted.

7.2.6.9 Gmac_Ip_SendFrame()

```
Gmac_Ip_StatusType Gmac_Ip_SendFrame (  
    uint8 Instance,  
    uint8 Ring,  
    const Gmac_Ip_BufferType * Buff,  
    const Gmac_Ip_TxOptionsType * Options )
```

Sends an Ethernet frame.

This function sends an Ethernet frame, contained in the buffer received as parameter.

Note: Since the transmission of the frame is not complete when this function returns, the application must not change/alter/re-use the provided buffer until after a call to Gmac_Ip_GetTransmitStatus for the same buffer returns GMAC_STATUS_SUCCESS.

Important: The driver does not ensure synchronization between different threads trying to send a frame at the same time. This synchronization shall be implemented by the application.

Parameters

in	<i>instance</i>	Instance number
in	<i>ring</i>	The ring number
in	<i>buff</i>	The buffer containing the frame
in	<i>options</i>	Configuration options applicable to this buffer's transmission only. Can be NULL_PTR, if no special option is required.

Return values

<i>GMAC_STATUS_SUCCESS</i>	The frame was successfully enqueued for transmission.
----------------------------	---

Return values

<i>GMAC_STATUS_TX_QUEUE_FULL</i>	There is no available space for the frame in the queue.
----------------------------------	---

7.2.6.10 Gmac_Ip_ReadFrame()

```
Gmac_Ip_StatusType Gmac_Ip_ReadFrame (
    uint8 Instance,
    uint8 Ring,
    Gmac_Ip_BufferType * Buff,
    Gmac_Ip_RxInfoType * Info )
```

Reads a received Ethernet frame.

This function reads the first received Ethernet frame in the Rx queue. The buffer received as parameter will be updated by the driver and the .data field will point to a memory area containing the frame data.

Note: Once the application finished processing the buffer, it could be reused by the driver for further receptions by invoking Gmac_Ip_ProvideRxBuff.

Important: The driver does not ensure synchronization between different threads trying to read a frame at the same time. This synchronization shall be implemented by the application.

Parameters

in	<i>instance</i>	Instance number
in	<i>ring</i>	The ring number
out	<i>buff</i>	The buffer containing the frame
out	<i>info</i>	Enhanced information related to the data contained by this receive buffer. If this information is not needed, this parameter should be NULL_PTR.

Return values

<i>GMAC_STATUS_SUCCESS</i>	A frame was successfully read.
<i>GMAC_STATUS_RX_QUEUE_EMPTY</i>	There is no available frame in the queue.

7.2.6.11 Gmac_Ip_ProvideRxBuff()

```
void Gmac_Ip_ProvideRxBuff (
    uint8 Instance,
    uint8 Ring,
    const Gmac_Ip_BufferType * Buff )
```


Provides a receive buffer to be used by the driver for reception.

This function provides a buffer which can further be used by the reception mechanism in order to store the received data.

Note: The application can either provide a buffer previously obtained in a `Gmac_Ip_ReadFrame` call (when it is no longer needed after being fully processed), or allocate a new buffer. The former approach is recommended as it has a simpler usage model and re-uses the same initial memory range for the entire driver lifetime operation. The later approach could provide more flexibility, but since it involves constant memory free/alloc operations it is only recommended with an efficient pool-based memory allocator.

Important: The driver does not ensure synchronization between different threads trying to provide a buffer at the same time. This synchronization shall be implemented by the application.

Important: The application is responsible for providing one Rx buffer for every frame it receives, otherwise the reception ring can fill-up, affecting further reception.

Usage example:

```
stat = Gmac_Ip_ReadFrame(INST_GMAC1, 0U, &rxBuff);

if (stat == GMAC_STATUS_SUCCESS) { process_buffer(&rxBuff); Gmac_Ip_ProvideRxBuff(INST_GMAC1,
0U, &rxBuff); }
```

Parameters

in	<i>instance</i>	Instance number
in	<i>ring</i>	The ring number
in	<i>buff</i>	The buffer to be added to the reception ring

7.2.6.12 Gmac_Ip_IsFrameAvailable()

```
boolean Gmac_Ip_IsFrameAvailable (
    uint8 Instance,
    uint8 Ring )
```

Checks if there are more frames available in the given queue.

This function takes a peek at the given Rx queue to check if there are more Ethernet frames to be received. Its intended usage is to provide this information without also extracting the frame as "Gmac_Ip_ReadFrame".

Parameters

in	<i>instance</i>	Instance number
in	<i>ring</i>	The ring number

Return values

<i>TRUE</i>	There is an available frame in the queue.
<i>FALSE</i>	There is no available frame in the queue.

7.2.6.13 Gmac_Ip_GetTransmitStatus()

```
Gmac_Ip_StatusType Gmac_Ip_GetTransmitStatus (
    uint8 Instance,
    uint8 Ring,
    const Gmac_Ip_BufferType * Buff,
    Gmac_Ip_TxInfoType * Info )
```

Checks if the transmission of a buffer is complete.

This function checks if the transmission of the given buffer is complete.

Note: If the buffer is not found in the Tx ring, the function will return `GMAC_STATUS_BUFF_NOT_FOUND`.

Parameters

in	<i>instance</i>	Instance number
in	<i>ring</i>	The ring number
in	<i>buff</i>	The transmit buffer for which the status shall be checked
out	<i>info</i>	Extended information related to the buffer. If this information is not needed, this parameter should be <code>NULL_PTR</code> .

Return values

<i>GMAC_STATUS_BUSY</i>	The frame is still enqueued for transmission.
<i>GMAC_STATUS_BUFF_NOT_FOUND</i>	The buffer was not found in the Tx queue.
<i>GMAC_STATUS_SUCCESS</i>	Otherwise.

7.2.6.14 Gmac_Ip_GetCounter()

```
uint32 Gmac_Ip_GetCounter (
    uint8 Instance,
    Gmac_Ip_CounterType Counter )
```

Gets statistics from the specified counter.

Parameters

in	<i>instance</i>	Instance number
in	<i>counter</i>	The counter to be read

Returns

The value of the requested counter

7.2.6.15 Gmac_Ip_EnableMDIO()

```
void Gmac_Ip_EnableMDIO (
    uint8 Instance,
    boolean MiiPreambleDisabled,
    uint32 ModuleClk )
```

Enables the MDIO interface.

Parameters

in	<i>instance</i>	Instance number
in	<i>miiPreambleDisabled</i>	Enables/disables prepending a preamble to the MII management frame.
in	<i>clock</i>	provide module

7.2.6.16 Gmac_Ip_MDIORead()

```
Gmac_Ip_StatusType Gmac_Ip_MDIORead (
    uint8 Instance,
    uint8 PhyAddr,
    uint8 PhyReg,
    uint16 * Data,
    uint32 TimeoutMs )
```

Reads the selected register of the PHY.

Parameters

in	<i>instance</i>	Instance number
in	<i>phyAddr</i>	PHY device address
in	<i>phyReg</i>	PHY register address
out	<i>data</i>	Data read from the PHY
in	<i>timeoutMs</i>	Timeout for the read operation (in milliseconds)

Return values

<i>GMAC_STATUS_SUCCESS</i>	The operation completed successfully.
<i>GMAC_STATUS_TIMEOUT</i>	The specified timeout expired before completing the operation.

7.2.6.17 Gmac_Ip_MDIOWrite()

```
Gmac_Ip_StatusType Gmac_Ip_MDIOWrite (
    uint8 Instance,
    uint8 PhyAddr,
    uint8 PhyReg,
    uint16 Data,
    uint32 TimeoutMs )
```

Writes the selected register of the PHY.

Parameters

in	<i>instance</i>	Instance number
in	<i>phyAddr</i>	PHY device address
in	<i>phyReg</i>	PHY register address
in	<i>data</i>	Data to be written in the specified register of the PHY
in	<i>timeoutMs</i>	Timeout for the write operation (in milliseconds)

Return values

<i>GMAC_STATUS_SUCCESS</i>	The operation completed successfully.
<i>GMAC_STATUS_ERROR</i>	The GMAC Mdio wasn't enabled beforehand.
<i>GMAC_STATUS_TIMEOUT</i>	The specified timeout expired before completing the operation.

7.2.6.18 Gmac_Ip_MDIOReadMMD()

```
Gmac_Ip_StatusType Gmac_Ip_MDIOReadMMD (
    uint8 Instance,
    uint8 PhyAddr,
    uint8 Mmd,
    uint16 PhyReg,
    uint16 * Data,
    uint32 TimeoutMs )
```

Reads a register of the specified MMD in a PHY device.

Parameters

in	<i>instance</i>	Instance number
in	<i>phyAddr</i>	PHY device address
in	<i>mmd</i>	The MMD index of the target register
in	<i>phyReg</i>	PHY register address
out	<i>data</i>	Data read from the PHY
in	<i>timeoutMs</i>	Timeout for the read operation (in milliseconds)

Return values

<i>GMAC_STATUS_SUCCESS</i>	The operation completed successfully.
<i>GMAC_STATUS_ERROR</i>	The GMAC Mdio wasn't enabled beforehand.
<i>GMAC_STATUS_TIMEOUT</i>	The specified timeout expired before completing the operation.

7.2.6.19 Gmac_Ip_MDIOWriteMMD()

```
Gmac_Ip_StatusType Gmac_Ip_MDIOWriteMMD (
    uint8 Instance,
    uint8 PhyAddr,
    uint8 Mmd,
    uint16 PhyReg,
    uint16 Data,
    uint32 TimeoutMs )
```

Writes a register of the specified MMD in a PHY device.

Parameters

in	<i>instance</i>	Instance number
in	<i>phyAddr</i>	PHY device address
in	<i>mmd</i>	The MMD index of the target register
in	<i>phyReg</i>	PHY register address
in	<i>data</i>	Data to be written in the specified register of the PHY
in	<i>timeoutMs</i>	Timeout for the write operation (in milliseconds)

Return values

<i>GMAC_STATUS_SUCCESS</i>	The operation completed successfully.
<i>GMAC_STATUS_TIMEOUT</i>	The specified timeout expired before completing the operation.

7.2.6.20 Gmac_Ip_SetMacAddr()

```
void Gmac_Ip_SetMacAddr (
    uint8 Instance,
    const uint8 * MacAddr )
```

Configures the physical address of the MAC.

Parameters

in	<i>instance</i>	Instance number
in	<i>macAddr</i>	The MAC address to be configured

7.2.6.21 Gmac_Ip_GetMacAddr()

```
void Gmac_Ip_GetMacAddr (
    uint8 Instance,
    uint8 * MacAddr )
```

Gets the physical address of the MAC.

Parameters

in	<i>instance</i>	Instance number
out	<i>macAddr</i>	The physical address of the MAC

7.2.6.22 Gmac_Ip_TxTimeAwareShaperInit()

```
Gmac_Ip_StatusType Gmac_Ip_TxTimeAwareShaperInit (
    uint8 Instance,
    const Gmac_CtrlConfigType * Config )
```

Initialize time aware shaper.

Parameters

in	<i>instance</i>	Instance number
in	<i>Parameters</i>	of controller will be used. return GMAC_STATUS_SUCCESS write/read to Gate control registers successfully return GMAC_STATUS_TIMEOUT write/read to Gate control registers failed

7.2.6.23 Gmac_Ip_GetInterruptFlags()

```
uint32 Gmac_Ip_GetInterruptFlags (
    uint8 Instance )
```

Gets a mask of the common interrupt events which occurred.

Parameters

in	<i>instance</i>	Instance number
----	-----------------	-----------------

Returns

A logical OR of "Gmac_Ip_InterruptType"

7.2.6.24 Gmac_Ip_GetChInterruptFlags()

```
uint32 Gmac_Ip_GetChInterruptFlags (
    uint8 Instance,
    uint8 Channel )
```

Gets a mask of the interrupt events associated to a channel which occurred.

Parameters

in	<i>instance</i>	Instance number
in	<i>channel</i>	Channel number

Returns

A logical OR of "Gmac_Ip_ChInterruptType"

7.2.6.25 Gmac_Ip_SetBroadcastForwardAll()

```
void Gmac_Ip_SetBroadcastForwardAll (
    uint8 Instance,
    boolean Enable )
```

Enables/Disables forwarding of the broadcast traffic.

Parameters

in	<i>instance</i>	Instance number
in	<i>enable</i>	If true, the application will receive all the broadcast traffic; if false, it stops forwarding this kind of traffic.

7.2.6.26 Gmac_Ip_SetMulticastForwardAll()

```
void Gmac_Ip_SetMulticastForwardAll (
    uint8 Instance,
    boolean Enable )
```

Enables/Disables forwarding of the multicast traffic, irrespective of the destination MAC address.

Parameters

in	<i>instance</i>	Instance number
in	<i>enable</i>	If true, the application will receive all the multicast traffic; if false, stops forwarding this kind of traffic.

7.2.6.27 Gmac_Ip_SetUnicastHashFilter()

```
void Gmac_Ip_SetUnicastHashFilter (
    uint8 Instance,
    boolean Enable )
```

Enables/Disables hash filtering for unicast traffic.

Note: When the hash filter is disabled, the addresses are passed through the perfect filter. By default, the perfect filter contains the hardware address configured when the MAC module is initialized. This address is always used for destination address filtering. Optionally, a second filter can be enabled by the means of [Gmac_Ip_SetAddrPerfectFilter\(\)](#) function.

Parameters

in	<i>instance</i>	Instance number
in	<i>enable</i>	If true, the hash filtering for unicast destination addresses is enabled; if false, the hash filtering for such traffic is disabled.

7.2.6.28 Gmac_Ip_SetMulticastHashFilter()

```
void Gmac_Ip_SetMulticastHashFilter (
    uint8 Instance,
    boolean Enable )
```

Enables/Disables hash filtering for multicast traffic.

Note: When the hash filter is disabled, the addresses are passed through the perfect filter. By default, the perfect filter contains the hardware address configured when the MAC module is initialized. This address is always used for destination address filtering. Optionally, a second filter can be enabled by the means of [Gmac_Ip_SetAddrPerfectFilter\(\)](#) function.

Parameters

in	<i>instance</i>	Instance number
in	<i>enable</i>	If true, the hash filtering for multicast destination addresses is enabled; if false, the hash filtering for such traffic is disabled.

7.2.6.29 Gmac_Ip_SetHashOrPerfectFilter()

```
void Gmac_Ip_SetHashOrPerfectFilter (
    uint8 Instance,
    boolean Enable )
```

Enables/Disables either hash or perfect filters. If the destination address of an incoming packet passes either the perfect filter or the hash filter, the packet is forwarded to the application.

Note: When the hash filter is disabled, the addresses are passed through the perfect filter. By default, the perfect filter contains the hardware address configured when the MAC module is initialized. This address is always used for destination address filtering. Optionally, a second filter can be enabled by the means of [Gmac_Ip_SetAddrPerfectFilter\(\)](#) function.

Parameters

in	<i>instance</i>	Instance number
in	<i>enable</i>	If true, the hash filtering and perfect filtering for destination addresses are enabled; if false, only one filtering option is enabled, depending on unicast/multicast hash filter settings.

7.2.6.30 Gmac_Ip_AddDstAddrToHashFilter()

```
void Gmac_Ip_AddDstAddrToHashFilter (
    uint8 Instance,
    const uint8 * MacAddr )
```

Adds a hardware address to the hash filter. The destination address of an incoming packet is passed through CRC logic and then compared to the entries in the hash table.

Parameters

in	<i>instance</i>	Instance number
in	<i>macAddr</i>	The physical address

7.2.6.31 Gmac_Ip_RemoveDstAddrFromHashFilter()

```
void Gmac_Ip_RemoveDstAddrFromHashFilter (
    uint8 Instance,
    const uint8 * MacAddr )
```

Removes a hardware address from the hash filter. The destination address of an incoming packet is passed through CRC logic and then compared to the entries in the hash table.

Parameters

in	<i>instance</i>	Instance number
in	<i>macAddr</i>	The physical address

7.2.6.32 Gmac_Ip_SetAddrPerfectFilter()

```
void Gmac_Ip_SetAddrPerfectFilter (
    uint8 Instance,
    const uint8 * MacAddr,
    const uint8 Mask,
    const Gmac_Ip_MacAddrFilterType AddrType,
    boolean Enable )
```

Sets the second destination/source address perfect filter.

Note: By default, the perfect filter contains the hardware address configured when the MAC module is initialized. This address is always used for destination address filtering. Optionally, a second filter can be enabled by the means of this function. This filter allows the selection of destination or source address checking and group filtering. Only one address can be configured and it's respective mask for group filtering. The mask is 6 bits long and is used for comparing each of the MAC Address bytes. When set high, the MAC does not compare the corresponding byte of received destination address or source address with the configured MAC address. You can filter a group of addresses by masking one or more bytes of the address. When configuring the source address inverse filter, all the traffic that matches the configured source address will be rejected and the traffic that does not match the configured source address is accepted. The source address filter (either normal or inverse) result and the destination address filter result are AND'ed to decide whether the packet to be accepted. This means that the packet is dropped if either filter fails. The packet is accepted only if the packet passes both filters.

Parameters

in	<i>instance</i>	Instance number
in	<i>macAddr</i>	The physical address
in	<i>mask</i>	Group filter mask
in	<i>addrType</i>	Selects between destination address filter, source address filter or source address inverse filter
in	<i>enable</i>	If true, the application will receive all the traffic matching the selected filter; if false, it stops forwarding this kind of traffic.

7.2.6.33 Gmac_Ip_SetArpOffloading()

```
void Gmac_Ip_SetArpOffloading (
    uint8 Instance,
    const uint8 * DstAddr,
    boolean Enable )
```

Enables/Disables ARP offloading.

Parameters

in	<i>instance</i>	Instance number
in	<i>dstAddr</i>	The IPv4 destination address used for ARP offloading
in	<i>enable</i>	If true, the MAC can recognize an incoming ARP request and schedules the ARP response for transmission. It also forwards the ARP packet to the application; if false, the MAC receiver does not recognize any ARP packet and indicates them as Type frame.

7.2.6.34 Gmac_Ip_EnableVlan()

```
void Gmac_Ip_EnableVlan (
    uint8 Instance,
    const Gmac_Ip_VlanConfigType * VlanConfig )
```

Enable and set VLAN control for transmitted and received packets.

Parameters

in	<i>instance</i>	Instance number
in	<i>vlanConfig</i>	VLAN configuration structure

7.2.6.35 Gmac_Ip_SetTxOuterVlanTagForInsertion()

```
Gmac_Ip_StatusType Gmac_Ip_SetTxOuterVlanTagForInsertion (
    uint8 Instance,
    uint8 Ring,
    Gmac_Ip_VlanType VlanType,
    uint16 VlanTag )
```

Sets outer VLAN type and tag to be inserted by a particular transmission ring.

Parameters

in	<i>instance</i>	Instance number
in	<i>ring</i>	Ring number
in	<i>vlanType</i>	VLAN type (S-VLAN or C-VLAN) to be used
in	<i>vlanTag</i>	Value for the VLAN tag to be used

Return values

<i>GMAC_STATUS_SUCCESS</i>	The ring was successfully configured.
<i>GMAC_STATUS_BUSY</i>	The resource is busy and the ring could not be configured.
<i>GMAC_STATUS_TIMEOUT</i>	The ring configuration could not complete before expiration of timeout.
<i>GMAC_STATUS_UNSUPPORTED</i>	The function was called when VLAN replacement option was configured.

7.2.6.36 Gmac_Ip_SetTxOuterVlanTagForReplacement()

```
Gmac_Ip_StatusType Gmac_Ip_SetTxOuterVlanTagForReplacement (
    uint8 Instance,
    Gmac_Ip_VlanType VlanType,
    uint16 VlanTag )
```

Sets outer VLAN tag to be replaced for all transmission rings.

Parameters

in	<i>instance</i>	Instance number
in	<i>vlanType</i>	VLAN type (S-VLAN or C-VLAN) to be used
in	<i>vlanTag</i>	Value for the VLAN tag to be used

Return values

<i>GMAC_STATUS_SUCCESS</i>	The rings were successfully configured.
<i>GMAC_STATUS_UNSUPPORTED</i>	The function was called when VLAN insertion option was configured.

7.2.6.37 Gmac_Ip_SetTxInnerVlanTag()

```
void Gmac_Ip_SetTxInnerVlanTag (
    uint8 Instance,
    Gmac_Ip_VlanType VlanType,
    uint16 VlanTag )
```

Sets inner VLAN type and tag to be inserted/replaced on transmission.

Parameters

in	<i>instance</i>	Instance number
in	<i>vlanType</i>	VLAN type (S-VLAN or C-VLAN) to be used
in	<i>vlanTag</i>	Value for the VLAN tag to be used

7.2.6.38 Gmac_Ip_SetVlanTagRxFilter()

```
void Gmac_Ip_SetVlanTagRxFilter (
    uint8 Instance,
    boolean Enable,
    const Gmac_Ip_VlanRxFilterType * RxFilter )
```

Enables/disables and configures the Rx filters for VLAN tag.

Parameters

in	<i>instance</i>	Instance number
in	<i>enable</i>	Enables/Disables the Rx filter for VLAN tag
in	<i>rxFilter</i>	Rx filter options for VLAN tag comparison

Note

For the GMAC that have the ERR050109 the support for DisVlanTypeMatch, EnSvlanMatch and En12bit↵ Match will not be available

7.2.6.39 Gmac_Ip_AddVlanTagToRxFilter()

```
Gmac_Ip_StatusType Gmac_Ip_AddVlanTagToRxFilter (
    uint8 Instance,
```

```
uint8 FilterIdx,
uint16 VlanTag )
```

Adds a VLAN Tag to the Rx filter.

Parameters

in	<i>instance</i>	Instance number
in	<i>filterIdx</i>	Index of the VLAN filter
in	<i>vlanTag</i>	Vlan Tag to be added to the Rx filter

Return values

<i>GMAC_STATUS_SUCCESS</i>	The filter was successfully configured.
<i>GMAC_STATUS_TIMEOUT</i>	The filter configuration could not complete before expiration of timeout.

7.2.6.40 Gmac_Ip_ReadVlanTagRxFilter()

```
Gmac_Ip_StatusType Gmac_Ip_ReadVlanTagRxFilter (
    uint8 Instance,
    uint8 FilterIdx,
    uint16 * VlanTag )
```

Read a VLAN Tag Rx filter.

Parameters

in	<i>instance</i>	Instance number
in	<i>filterIdx</i>	Index of the VLAN filter
out	<i>vlanTag</i>	Vlan Tag configured for the selected Rx filter

Return values

<i>GMAC_STATUS_SUCCESS</i>	The filter was successfully read.
<i>GMAC_STATUS_TIMEOUT</i>	The filter could not be read before expiration of timeout.

7.2.6.41 Gmac_Ip_ClearVlanTagRxFilter()

```
Gmac_Ip_StatusType Gmac_Ip_ClearVlanTagRxFilter (
    uint8 Instance,
    uint8 FilterIdx )
```

Clear a VLAN Tag Rx filter.

Parameters

in	<i>instance</i>	Instance number
in	<i>filterIdx</i>	Index of the VLAN filter

Return values

<i>GMAC_STATUS_SUCCESS</i>	The filter was successfully cleared.
<i>GMAC_STATUS_TIMEOUT</i>	The filter could not be cleared before expiration of timeout.

7.2.6.42 Gmac_Ip_AddVlanTagToHashTable()

```
void Gmac_Ip_AddVlanTagToHashTable (
    uint8 Instance,
    uint16 VlanTag )
```

Adds a VLAN Tag to the Hash Table filter.

Parameters

in	<i>instance</i>	Instance number
in	<i>vlanTag</i>	Vlan Tag to be added to the hash table

7.2.6.43 Gmac_Ip_RemoveVlanTagFromHashTable()

```
void Gmac_Ip_RemoveVlanTagFromHashTable (
    uint8 Instance,
    uint16 VlanTag )
```

Removes a VLAN Tag from the hash table filter.

Parameters

in	<i>instance</i>	Instance number
in	<i>vlanTag</i>	Vlan Tag to be removed from the hash table

7.2.6.44 Gmac_Ip_InitSysTime()

```
void Gmac_Ip_InitSysTime (
    uint8 Instance,
    const Gmac_Ip_SysTimeConfigType * SysTimeConfig )
```

Initialize system time.

Parameters

in	<i>instance</i>	Instance number
in	<i>sysTimeConfig</i>	Pointer to a structure representing the configuration of the system time

7.2.6.45 Gmac_Ip_SetSysTimeCorr()

```
Gmac_Ip_StatusType Gmac_Ip_SetSysTimeCorr (
    uint8 Instance,
    Gmac_Ip_SysTimeCorrOffsetType Offset,
    uint32 SecondsUpdate,
    uint32 NanoSecondsUpdate )
```

Set system time correction.

Parameters

in	<i>instance</i>	Instance number
in	<i>offset</i>	Chooses between negative and positive correction
in	<i>secondsUpdate</i>	The seconds part of the update
in	<i>nanosecondsUpdate</i>	The nanoseconds part of the update

Return values

<i>GMAC_STATUS_SUCCESS</i>	The correction was set with no error.
<i>GMAC_STATUS_TIMEOUT</i>	The correction could not be set before expiration of timeout.

7.2.6.46 Gmac_Ip_SetRateRatioCorr()

```
Gmac_Ip_StatusType Gmac_Ip_SetRateRatioCorr (
    uint8 Instance,
    float64 RateRatio )
```

Set system time correction.

Module Documentation

Parameters

in	<i>instance</i>	Instance number
in	<i>rateRatio</i>	Rate ratio is used to sync

Return values

<i>GMAC_STATUS_SUCCESS</i>	The correction was set with no error.
<i>GMAC_STATUS_TIMEOUT</i>	The correction could not be set before expiration of timeout.

7.2.6.47 Gmac_Ip_GetSysTime()

```
void Gmac_Ip_GetSysTime (
    uint8 Instance,
    Gmac_Ip_TimestampType * Timestamp )
```

Gets the current system time.

Parameters

in	<i>instance</i>	Instance number
out	<i>timestamp</i>	Pointer to a structure representing the current system time

7.2.6.48 Gmac_Ip_SetTxSchedAlgo()

```
void Gmac_Ip_SetTxSchedAlgo (
    uint8 Instance,
    Gmac_Ip_TxSchedAlgoType SchedAlgo )
```

Sets the transmitter scheduling algorithm.

Parameters

in	<i>instance</i>	Instance number
in	<i>scheme</i>	Transmitter scheduling algorithm.

7.2.6.49 Gmac_Ip_SetTxRingWeight()

```
void Gmac_Ip_SetTxRingWeight (
    uint8 Instance,
    uint8 Ring,
    uint32 Weight )
```

Sets the weight (in WRR scheduling algorithm) for a particular Tx ring.

Note: If multiple rings are configured and the WRR scheduling algorithm is selected, the weight must be programmed with a non-zero value. The maximum value for the weight is 0x64.

Parameters

in	<i>instance</i>	Instance number
in	<i>ring</i>	Tx ring
in	<i>weight</i>	Tx ring weight

7.2.6.50 Gmac_Ip_EnableTxStoreAndForward()

```
void Gmac_Ip_EnableTxStoreAndForward (
    uint8 Instance,
    uint8 Ring )
```

Enables the store and forward feature on the transmit path.

When the store and forward feature is enabled, the MAC starts to transmit data only when a complete packet is stored in the Transmit Queue.

Parameters

in	<i>instance</i>	Instance number
in	<i>ring</i>	Tx ring

7.2.6.51 Gmac_Ip_SetTxThreshold()

```
void Gmac_Ip_SetTxThreshold (
    uint8 Instance,
    uint8 Ring,
    Gmac_Ip_TxThresholdType ThresholdValue )
```

Sets transmit threshold levels.

Module Documentation

Parameters

in	<i>instance</i>	Instance number
in	<i>ring</i>	Tx ring
in	<i>thresholdValue</i>	The threshold value



Chapter 8

Data Structure Documentation

8.1 Gmac_Ip_ManagementInfo Struct Reference

Management Frame information.

```
#include <Gmac_Ip_Hw_Access.h>
```

8.1.1 Detailed Description

Management Frame information.

Definition at line 109 of file Gmac_Ip_Hw_Access.h.

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